

SAPIENZA
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Integer Linear Programming

Exercises 1 — Models, bounds and formulations

Worked exercises on chapters 1–5, with complete solutions. Each exercise cites the catalogue problem it uses.

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Online version and downloadable material:
fabiofurini.github.io/integer-linear-programming/
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Part II

Strengthening the formulation

Chapter 5

Cover inequalities

5.1 Cover separation

Exercise 5.1 — Cover inequalities ($P1$)

Consider a binary problem with one constraint $\sum_j w_j x_j \leq c$, with positive integer coefficients w_j and right-hand side $c = 8$, on the binary variables indexed by the set $\mathscr{J} = \{1, 2, 3, 4\}$ of $n = 4$ variables with the following objective coefficients and constraint coefficients:

j	1	2	3	4
p_j	40	42	25	12
w_j	4	7	5	3

The associated LP is:

$$
\begin{array}{rrcrcrcrcl}
\max & 40\,x_1 & + & 42\,x_2 & + & 25\,x_3 & + & 12\,x_4 & & \\
\text{subject to} & 4\,x_1 & + & 7\,x_2 & + & 5\,x_3 & + & 3\,x_4 & \leq & 8 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}
$$

An optimal solution $\tilde{\boldsymbol{x}}$ of LP and its value are:

$$
\tilde{x}_1 = 1, \quad \tilde{x}_2 = \frac{4}{7}, \quad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 0 \qquad \text{and} \qquad z(\text{LP}) = 64.
$$

Questions

1. List all the cover inequalities and identify the minimal covers.
2. For the point $\tilde{\boldsymbol{x}}$, write the separation problem ILP_{CSP} and its complementary problem $\text{ILP}^2_{\text{CSP}}$.

3. Apply the greedy heuristic to $\text{ILP}^2_{\text{CSP}}$ and, if possible, find a violated cover inequality.
4. Check whether the greedy solution is optimal.

A subset of indices $C \subseteq \mathscr{I}$ with $\sum_{j \in C} w_j > c$ is a **cover**, and the linear inequality

$$\sum_{j \in C} x_j \leq |C| - 1$$

is the **cover inequality** associated with C. Given a point $\tilde{\boldsymbol{x}}$, a cover inequality violated by $\tilde{\boldsymbol{x}}$ exists if and only if $z(\text{ILP}_{\text{CSP}}) < 1$, where

$$(\text{ILP}_{\text{CSP}}) \qquad \min \Big\{ \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j)\, z_j : \sum_{j \in \mathscr{I}} w_j\, z_j \geq c + 1,\ z_j \in \{0,1\}\ \forall j \in \mathscr{I} \Big\}.$$

Solution

Separation of the cover inequalities violated by $\tilde{\boldsymbol{x}}$.

- **Cover inequalities**
  The covers are the subsets of indices whose total coefficient exceeds $c = 8$:

C	$\sum_{j \in C} w_j$	$\lvert C \rvert$	Cover inequality	Minimal
$\{1,2\}$	11	2	$x_1 + x_2 \leq 1$	yes
$\{1,3\}$	9	2	$x_1 + x_3 \leq 1$	yes
$\{2,3\}$	12	2	$x_2 + x_3 \leq 1$	yes
$\{2,4\}$	10	2	$x_2 + x_4 \leq 1$	yes
$\{1,2,3\}$	16	3	$x_1 + x_2 + x_3 \leq 2$	no
$\{1,2,4\}$	14	3	$x_1 + x_2 + x_4 \leq 2$	no
$\{1,3,4\}$	12	3	$x_1 + x_3 + x_4 \leq 2$	no
$\{2,3,4\}$	15	3	$x_2 + x_3 + x_4 \leq 2$	no
$\{1,2,3,4\}$	19	4	$x_1 + x_2 + x_3 + x_4 \leq 3$	no

  There are 9 covers, of which 4 are minimal. The cover inequalities of the non-minimal covers are implied by those of the minimal covers and by the bounds $x_j \leq 1$.

- **Separation problem**
  The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(0, \frac{3}{7}, 1, 1\right)$, and the separation problem ILP_{CSP} is:

$$\begin{array}{lccccccccc} \min & & & \frac{3}{7}\, z_2 & + & z_3 & + & z_4 & & \\ \text{subject to} & 4\, z_1 & + & 7\, z_2 & + & 5\, z_3 & + & 3\, z_4 & \geq & 9 \\ & z_1, & & z_2, & & z_3, & & z_4 & \in & \{0,1\} \end{array}$$

  With $y_j = 1 - z_j$, the constant term and the right-hand side of the complementary

problem are

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j = \frac{17}{7}, \qquad \sum_{j \in \mathscr{I}} w_j - (c+1) = 19 - 9 = 10,$$

and the complementary problem $\text{ILP}^2_{\text{CSP}}$ is:

$$\begin{array}{lrcrcrcrcl} \max & & & \frac{3}{7}\,y_2 & + & y_3 & + & y_4 & & \\ \text{subject to} & 4\,y_1 & + & 7\,y_2 & + & 5\,y_3 & + & 3\,y_4 & \leq & 10 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

with $z(\text{ILP}_{\text{CSP}}) = \frac{17}{7} - z(\text{ILP}^2_{\text{CSP}})$.

- **Greedy heuristic**
  The indices are scanned by non-increasing ratio q_j/w_j, starting from the residual right-hand side 10:

j	q_j	w_j	q_j/w_j	Decision	Residual
4	1	3	$\frac{1}{3}$	take	7
3	1	5	$\frac{1}{5}$	take	2
2	$\frac{3}{7}$	7	$\frac{3}{49}$	skip	2
1	0	4	0	$q_j = 0$	2

  The greedy solution is $\bar{\boldsymbol{y}} = (0,0,1,1)$, with value 2. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{1,2\}$, of total coefficient $11 > 8$ and cost $\frac{17}{7} - 2 = \frac{3}{7} < 1$. Hence the cover inequality

$$x_1 + x_2 \leq 1$$

  is violated by $\tilde{\boldsymbol{x}}$: $1 + \frac{4}{7} - 1 = \frac{4}{7}$.

- **Optimality check**
  Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{2,3,4\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 10 are: $\{2\}$ (value $\frac{3}{7}$), $\{3\}$ (value 1), $\{4\}$ (value 1), $\{2,4\}$ (value $\frac{10}{7}$), $\{3,4\}$ (value 2). Hence $z(\text{ILP}^2_{\text{CSP}}) = 2$, the greedy solution is optimal, $z(\text{ILP}_{\text{CSP}}) = \frac{3}{7} < 1$, and the maximum violation of a cover inequality is $\frac{4}{7}$.

Exercise 5.2 — Cover inequalities ($P2$)

Consider a binary problem with one constraint $\sum_j w_j x_j \leq c$, with positive integer coefficients w_j and right-hand side $c = 15$, on the binary variables indexed by the set $\mathscr{I} = \{1,2,3,4\}$ of $n = 4$ variables with the following objective coefficients and constraint coefficients:

j	1	2	3	4
p_j	24	17	12	6
w_j	10	8	6	5

The associated LP is:

$$\begin{array}{rrcrcrcrcl} \max & 24\,x_1 & + & 17\,x_2 & + & 12\,x_3 & + & 6\,x_4 & & \\ \text{subject to} & 10\,x_1 & + & 8\,x_2 & + & 6\,x_3 & + & 5\,x_4 & \leq & 15 \\ & x_1 & & & & & & & \leq & 1 \\ & & & x_2 & & & & & \leq & 1 \\ & & & & & x_3 & & & \leq & 1 \\ & & & & & & & x_4 & \leq & 1 \\ & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0 \end{array}$$

An optimal solution $\tilde{\boldsymbol{x}}$ of LP and its value are:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = \frac{5}{8}, \quad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 0 \qquad \text{and} \qquad z(\text{LP}) = \frac{277}{8}.$$

Questions

1. List all the cover inequalities and identify the minimal covers.
2. For the point $\tilde{\boldsymbol{x}}$, write the separation problem ILP_{CSP} and its complementary problem $\text{ILP}^2_{\text{CSP}}$.
3. Apply the greedy heuristic to $\text{ILP}^2_{\text{CSP}}$ and, if possible, find a violated cover inequality.
4. Check whether the greedy solution is optimal.

A subset of indices $C \subseteq \mathscr{I}$ with $\sum_{j \in C} w_j > c$ is a **cover**, and the linear inequality

$$\sum_{j \in C} x_j \leq |C| - 1$$

is the **cover inequality** associated with C. Given a point $\tilde{\boldsymbol{x}}$, a cover inequality violated by $\tilde{\boldsymbol{x}}$ exists if and only if $z(\text{ILP}_{\text{CSP}}) < 1$, where

$$(\text{ILP}_{\text{CSP}}) \qquad \min \left\{ \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j)\, z_j : \sum_{j \in \mathscr{I}} w_j\, z_j \geq c + 1,\ z_j \in \{0,1\}\ \forall j \in \mathscr{I} \right\}.$$

Solution

Separation of the cover inequalities violated by $\tilde{\boldsymbol{x}}$.

- **Cover inequalities**
  The covers are the subsets of indices whose total coefficient exceeds $c = 15$:

C	$\sum_{j\in C} w_j$	$\lvert C\rvert$	Cover inequality	Minimal
$\{1,2\}$	18	2	$x_1 + x_2 \le 1$	yes
$\{1,3\}$	16	2	$x_1 + x_3 \le 1$	yes
$\{1,2,3\}$	24	3	$x_1 + x_2 + x_3 \le 2$	no
$\{1,2,4\}$	23	3	$x_1 + x_2 + x_4 \le 2$	no
$\{1,3,4\}$	21	3	$x_1 + x_3 + x_4 \le 2$	no
$\{2,3,4\}$	19	3	$x_2 + x_3 + x_4 \le 2$	yes
$\{1,2,3,4\}$	29	4	$x_1 + x_2 + x_3 + x_4 \le 3$	no

There are 7 covers, of which 3 are minimal. The cover inequalities of the non-minimal covers are implied by those of the minimal covers and by the bounds $x_j \le 1$.

- **Separation problem**
  The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(0, \frac{3}{8}, 1, 1\right)$, and the separation problem ILP_{CSP} is:

$$\begin{array}{rccccccccc} \min & & & \frac{3}{8}\,z_2 & + & z_3 & + & z_4 & & \\ \text{subject to} & 10\,z_1 & + & 8\,z_2 & + & 6\,z_3 & + & 5\,z_4 & \ge & 16 \\ & z_1, & & z_2, & & z_3, & & z_4 & \in & \{0,1\} \end{array}$$

  With $y_j = 1 - z_j$, the constant term and the right-hand side of the complementary problem are

$$n - \sum_{j\in\mathscr{I}} \tilde{x}_j = \frac{19}{8}, \qquad \sum_{j\in\mathscr{I}} w_j - (c+1) = 29 - 16 = 13,$$

  and the complementary problem $\text{ILP}^2_{\text{CSP}}$ is:

$$\begin{array}{rccccccccc} \max & & & \frac{3}{8}\,y_2 & + & y_3 & + & y_4 & & \\ \text{subject to} & 10\,y_1 & + & 8\,y_2 & + & 6\,y_3 & + & 5\,y_4 & \le & 13 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

  with $z(\text{ILP}_{\text{CSP}}) = \frac{19}{8} - z(\text{ILP}^2_{\text{CSP}})$.

- **Greedy heuristic**
  The indices are scanned by non-increasing ratio q_j/w_j, starting from the residual right-hand side 13:

j	q_j	w_j	q_j/w_j	Decision	Residual
4	1	5	$\frac{1}{5}$	take	8
3	1	6	$\frac{1}{6}$	take	2
2	$\frac{3}{8}$	8	$\frac{3}{64}$	skip	2
1	0	10	0	$q_j = 0$	2

The greedy solution is $\bar{\boldsymbol{y}} = (0, 0, 1, 1)$, with value 2. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{1, 2\}$, of total coefficient $18 > 15$ and cost $\frac{19}{8} - 2 = \frac{3}{8} < 1$. Hence the cover inequality

$$x_1 + x_2 \leq 1$$

is violated by $\tilde{\boldsymbol{x}}$: $1 + \frac{5}{8} - 1 = \frac{5}{8}$.

- **Optimality check**
  Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{2, 3, 4\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 13 are: $\{2\}$ (value $\frac{3}{8}$), $\{3\}$ (value 1), $\{4\}$ (value 1), $\{2, 4\}$ (value $\frac{11}{8}$), $\{3, 4\}$ (value 2). Hence $z(\text{ILP}^2_{\text{CSP}}) = 2$, the greedy solution is optimal, $z(\text{ILP}_{\text{CSP}}) = \frac{3}{8} < 1$, and the maximum violation of a cover inequality is $\frac{5}{8}$.

Exercise 5.3 — Cover inequalities ($P3$)

Consider a binary problem with one constraint $\sum_j w_j x_j \leq c$, with positive integer coefficients w_j and right-hand side $c = 12$, on the binary variables indexed by the set $\mathscr{I} = \{1, 2, 3, 4\}$ of $n = 4$ variables with the following objective coefficients and constraint coefficients:

j	1	2	3	4
p_j	38	14	44	19
w_j	5	2	10	6

The associated LP is:

$$\begin{array}{rrcrcrcrcl}
\max & 38\,x_1 & + & 14\,x_2 & + & 44\,x_3 & + & 19\,x_4 & & \\
\text{subject to} & 5\,x_1 & + & 2\,x_2 & + & 10\,x_3 & + & 6\,x_4 & \leq & 12 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}$$

An optimal solution $\tilde{\boldsymbol{x}}$ of LP and its value are:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_3 = \frac{1}{2}, \quad \tilde{x}_4 = 0 \qquad \text{and} \qquad z(\text{LP}) = 74.$$

Questions

1. List all the cover inequalities and identify the minimal covers.
2. For the point $\tilde{\boldsymbol{x}}$, write the separation problem ILP_{CSP} and its complementary problem $\text{ILP}^2_{\text{CSP}}$.
3. Apply the greedy heuristic to $\text{ILP}^2_{\text{CSP}}$ and, if possible, find a violated cover inequality.

4. Check whether the greedy solution is optimal.

A subset of indices $C \subseteq \mathscr{I}$ with $\sum_{j \in C} w_j > c$ is a **cover**, and the linear inequality

$$\sum_{j \in C} x_j \leq |C| - 1$$

is the **cover inequality** associated with C. Given a point $\tilde{\boldsymbol{x}}$, a cover inequality violated by $\tilde{\boldsymbol{x}}$ exists if and only if $z(\text{ILP}_{\text{CSP}}) < 1$, where

$$(\text{ILP}_{\text{CSP}}) \qquad \min \Big\{ \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j)\, z_j : \sum_{j \in \mathscr{I}} w_j\, z_j \geq c + 1,\ z_j \in \{0,1\}\ \forall j \in \mathscr{I} \Big\}.$$

Solution

Separation of the cover inequalities violated by $\tilde{\boldsymbol{x}}$.

- **Cover inequalities**
  The covers are the subsets of indices whose total coefficient exceeds $c = 12$:

C	$\sum_{j \in C} w_j$	$\lvert C \rvert$	Cover inequality	Minimal
$\{1,3\}$	15	2	$x_1 + x_3 \leq 1$	yes
$\{3,4\}$	16	2	$x_3 + x_4 \leq 1$	yes
$\{1,2,3\}$	17	3	$x_1 + x_2 + x_3 \leq 2$	no
$\{1,2,4\}$	13	3	$x_1 + x_2 + x_4 \leq 2$	yes
$\{1,3,4\}$	21	3	$x_1 + x_3 + x_4 \leq 2$	no
$\{2,3,4\}$	18	3	$x_2 + x_3 + x_4 \leq 2$	no
$\{1,2,3,4\}$	23	4	$x_1 + x_2 + x_3 + x_4 \leq 3$	no

  There are 7 covers, of which 3 are minimal. The cover inequalities of the non-minimal covers are implied by those of the minimal covers and by the bounds $x_j \leq 1$.

- **Separation problem**
  The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(0, 0, \frac{1}{2}, 1\right)$, and the separation problem ILP_{CSP} is:

$$\begin{array}{lccccccccc} \min & & & & & \frac{1}{2}\, z_3 & + & z_4 & & \\ \text{subject to} & 5\, z_1 & + & 2\, z_2 & + & 10\, z_3 & + & 6\, z_4 & \geq & 13 \\ & z_1, & & z_2, & & z_3, & & z_4 & \in & \{0,1\} \end{array}$$

  With $y_j = 1 - z_j$, the constant term and the right-hand side of the complementary problem are

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j = \frac{3}{2}, \qquad \sum_{j \in \mathscr{I}} w_j - (c+1) = 23 - 13 = 10,$$

  and the complementary problem $\text{ILP}^2_{\text{CSP}}$ is:

$$\begin{array}{lccccccccc} \max & & & & & \frac{1}{2}\,y_3 & + & y_4 & & \\ \text{subject to} & 5\,y_1 & + & 2\,y_2 & + & 10\,y_3 & + & 6\,y_4 & \leq & 10 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

with $z(\text{ILP}_{\text{CSP}}) = \frac{3}{2} - z(\text{ILP}^2_{\text{CSP}})$.

- **Greedy heuristic**
  The indices are scanned by non-increasing ratio q_j/w_j, starting from the residual right-hand side 10:

j	q_j	w_j	q_j/w_j	Decision	Residual
4	1	6	$\frac{1}{6}$	take	4
3	$\frac{1}{2}$	10	$\frac{1}{20}$	skip	4
1	0	5	0	$q_j = 0$	4
2	0	2	0	$q_j = 0$	4

  The greedy solution is $\bar{\boldsymbol{y}} = (0,0,0,1)$, with value 1. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{1,2,3\}$, of total coefficient $17 > 12$ and cost $\frac{3}{2} - 1 = \frac{1}{2} < 1$. Hence the cover inequality

$$x_1 + x_2 + x_3 \leq 2$$

  is violated by $\tilde{\boldsymbol{x}}$: $1 + 1 + \frac{1}{2} - 2 = \frac{1}{2}$.

- **Optimality check**
  Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{3,4\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 10 are: $\{3\}$ (value $\frac{1}{2}$), $\{4\}$ (value 1). Hence $z(\text{ILP}^2_{\text{CSP}}) = 1$, the greedy solution is optimal, $z(\text{ILP}_{\text{CSP}}) = \frac{1}{2} < 1$, and the maximum violation of a cover inequality is $\frac{1}{2}$.

  The cover $\bar{C}$ is not minimal: removing an index with $q_j = 0$ gives the minimal cover $\{1,3\}$, with the same cost $\frac{1}{2}$. Its cover inequality $x_1 + x_3 \leq 1$ has the same violation $\frac{1}{2}$ and is stronger.

Exercise 5.4 — Cover inequalities ($P4$)

Consider a binary problem with one constraint $\sum_j w_j x_j \leq c$, with positive integer coefficients w_j and right-hand side $c = 11$, on the binary variables indexed by the set $\mathscr{I} = \{1,2,3,4\}$ of $n = 4$ variables with the following objective coefficients and constraint coefficients:

j	1	2	3	4
p_j	34	40	31	12
w_j	5	9	9	5

The associated LP is:

$$
\begin{array}{rrcrcrcrcl}
\max & 34\,x_1 & + & 40\,x_2 & + & 31\,x_3 & + & 12\,x_4 & & \\
\text{subject to} & 5\,x_1 & + & 9\,x_2 & + & 9\,x_3 & + & 5\,x_4 & \leq & 11 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}
$$

An optimal solution $\tilde{\boldsymbol{x}}$ of LP and its value are:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = \frac{2}{3}, \quad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 0 \qquad \text{and} \qquad z(\text{LP}) = \frac{182}{3}.$$

Questions

1. List all the cover inequalities and identify the minimal covers.
2. For the point $\tilde{\boldsymbol{x}}$, write the separation problem ILP_{CSP} and its complementary problem $\text{ILP}^2_{\text{CSP}}$.
3. Apply the greedy heuristic to $\text{ILP}^2_{\text{CSP}}$ and, if possible, find a violated cover inequality.
4. Check whether the greedy solution is optimal.

A subset of indices $C \subseteq \mathscr{I}$ with $\sum_{j \in C} w_j > c$ is a **cover**, and the linear inequality

$$\sum_{j \in C} x_j \leq |C| - 1$$

is the **cover inequality** associated with C. Given a point $\tilde{\boldsymbol{x}}$, a cover inequality violated by $\tilde{\boldsymbol{x}}$ exists if and only if $z(\text{ILP}_{\text{CSP}}) < 1$, where

$$(\text{ILP}_{\text{CSP}}) \qquad \min \Big\{ \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j)\, z_j : \sum_{j \in \mathscr{I}} w_j\, z_j \geq c + 1,\ z_j \in \{0,1\}\ \forall j \in \mathscr{I} \Big\}.$$

Solution

Separation of the cover inequalities violated by $\tilde{\boldsymbol{x}}$.

- **Cover inequalities**
  The covers are the subsets of indices whose total coefficient exceeds $c = 11$:

C	$\sum_{j\in C} w_j$	$\lvert C\rvert$	Cover inequality	Minimal
$\{1,2\}$	14	2	$x_1 + x_2 \le 1$	yes
$\{1,3\}$	14	2	$x_1 + x_3 \le 1$	yes
$\{2,3\}$	18	2	$x_2 + x_3 \le 1$	yes
$\{2,4\}$	14	2	$x_2 + x_4 \le 1$	yes
$\{3,4\}$	14	2	$x_3 + x_4 \le 1$	yes
$\{1,2,3\}$	23	3	$x_1 + x_2 + x_3 \le 2$	no
$\{1,2,4\}$	19	3	$x_1 + x_2 + x_4 \le 2$	no
$\{1,3,4\}$	19	3	$x_1 + x_3 + x_4 \le 2$	no
$\{2,3,4\}$	23	3	$x_2 + x_3 + x_4 \le 2$	no
$\{1,2,3,4\}$	28	4	$x_1 + x_2 + x_3 + x_4 \le 3$	no

There are 10 covers, of which 5 are minimal. The cover inequalities of the non-minimal covers are implied by those of the minimal covers and by the bounds $x_j \le 1$.

- **Separation problem**
  The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(0, \frac{1}{3}, 1, 1\right)$, and the separation problem $\mathrm{ILP}_{\mathrm{CSP}}$ is:

$$\begin{array}{rccccccccc} \min & & & \frac{1}{3}\,z_2 & + & z_3 & + & z_4 & & \\ \text{subject to} & 5\,z_1 & + & 9\,z_2 & + & 9\,z_3 & + & 5\,z_4 & \ge & 12 \\ & z_1, & & z_2, & & z_3, & & z_4 & \in & \{0,1\} \end{array}$$

  With $y_j = 1 - z_j$, the constant term and the right-hand side of the complementary problem are

$$n - \sum_{j\in\mathscr{I}} \tilde{x}_j = \frac{7}{3}, \qquad \sum_{j\in\mathscr{I}} w_j - (c+1) = 28 - 12 = 16,$$

  and the complementary problem $\mathrm{ILP}^2_{\mathrm{CSP}}$ is:

$$\begin{array}{rccccccccc} \max & & & \frac{1}{3}\,y_2 & + & y_3 & + & y_4 & & \\ \text{subject to} & 5\,y_1 & + & 9\,y_2 & + & 9\,y_3 & + & 5\,y_4 & \le & 16 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

  with $z(\mathrm{ILP}_{\mathrm{CSP}}) = \frac{7}{3} - z(\mathrm{ILP}^2_{\mathrm{CSP}})$.

- **Greedy heuristic**
  The indices are scanned by non-increasing ratio q_j/w_j, starting from the residual right-hand side 16:

j	q_j	w_j	q_j/w_j	Decision	Residual
4	1	5	$\frac{1}{5}$	take	11
3	1	9	$\frac{1}{9}$	take	2
2	$\frac{1}{3}$	9	$\frac{1}{27}$	skip	2
1	0	5	0	$q_j = 0$	2

The greedy solution is $\bar{\boldsymbol{y}} = (0, 0, 1, 1)$, with value 2. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{1, 2\}$, of total coefficient $14 > 11$ and cost $\frac{7}{3} - 2 = \frac{1}{3} < 1$. Hence the cover inequality

$$x_1 + x_2 \leq 1$$

is violated by $\tilde{\boldsymbol{x}}$: $1 + \frac{2}{3} - 1 = \frac{2}{3}$.

- **Optimality check**
  Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{2, 3, 4\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 16 are: $\{2\}$ (value $\frac{1}{3}$), $\{3\}$ (value 1), $\{4\}$ (value 1), $\{2, 4\}$ (value $\frac{4}{3}$), $\{3, 4\}$ (value 2). Hence $z(\text{ILP}^2_{\text{CSP}}) = 2$, the greedy solution is optimal, $z(\text{ILP}_{\text{CSP}}) = \frac{1}{3} < 1$, and the maximum violation of a cover inequality is $\frac{2}{3}$.

Exercise 5.5 — Cover inequalities ($P5$)

Consider a binary problem with one constraint $\sum_j w_j x_j \leq c$, with positive integer coefficients w_j and right-hand side $c = 12$, on the binary variables indexed by the set $\mathscr{I} = \{1, 2, 3, 4\}$ of $n = 4$ variables with the following objective coefficients and constraint coefficients:

j	1	2	3	4
p_j	41	40	24	11
w_j	3	5	5	5

The associated LP is:

$$\begin{array}{rrcrcrcrcl}
\max & 41\,x_1 & + & 40\,x_2 & + & 24\,x_3 & + & 11\,x_4 & & \\
\text{subject to} & 3\,x_1 & + & 5\,x_2 & + & 5\,x_3 & + & 5\,x_4 & \leq & 12 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}$$

An optimal solution $\tilde{\boldsymbol{x}}$ of LP and its value are:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_3 = \frac{4}{5}, \quad \tilde{x}_4 = 0 \qquad \text{and} \qquad z(\text{LP}) = \frac{501}{5}.$$

Questions

1. List all the cover inequalities and identify the minimal covers.
2. For the point $\tilde{\boldsymbol{x}}$, write the separation problem ILP_{CSP} and its complementary problem $\text{ILP}^2_{\text{CSP}}$.
3. Apply the greedy heuristic to $\text{ILP}^2_{\text{CSP}}$ and, if possible, find a violated cover inequality.

4. Check whether the greedy solution is optimal.

A subset of indices $C \subseteq \mathscr{I}$ with $\sum_{j \in C} w_j > c$ is a **cover**, and the linear inequality

$$\sum_{j \in C} x_j \leq |C| - 1$$

is the **cover inequality** associated with C. Given a point $\tilde{\boldsymbol{x}}$, a cover inequality violated by $\tilde{\boldsymbol{x}}$ exists if and only if $z(\text{ILP}_{\text{CSP}}) < 1$, where

$$(\text{ILP}_{\text{CSP}}) \qquad \min \Big\{ \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j)\, z_j : \sum_{j \in \mathscr{I}} w_j\, z_j \geq c + 1,\ z_j \in \{0,1\}\ \forall j \in \mathscr{I} \Big\}.$$

Solution

Separation of the cover inequalities violated by $\tilde{\boldsymbol{x}}$.

- **Cover inequalities**
  The covers are the subsets of indices whose total coefficient exceeds $c = 12$:

C	$\sum_{j \in C} w_j$	$\lvert C \rvert$	Cover inequality	Minimal
$\{1,2,3\}$	13	3	$x_1 + x_2 + x_3 \leq 2$	yes
$\{1,2,4\}$	13	3	$x_1 + x_2 + x_4 \leq 2$	yes
$\{1,3,4\}$	13	3	$x_1 + x_3 + x_4 \leq 2$	yes
$\{2,3,4\}$	15	3	$x_2 + x_3 + x_4 \leq 2$	yes
$\{1,2,3,4\}$	18	4	$x_1 + x_2 + x_3 + x_4 \leq 3$	no

  There are 5 covers, of which 4 are minimal. The cover inequalities of the non-minimal covers are implied by those of the minimal covers and by the bounds $x_j \leq 1$.

- **Separation problem**
  The costs $q_j = 1 - \tilde{x}_j$ are $q = \big(0, 0, \frac{1}{5}, 1\big)$, and the separation problem ILP_{CSP} is:

$$\begin{array}{rccccccccc} \min & & & & & \frac{1}{5}\, z_3 & + & z_4 & & \\ \text{subject to} & 3\, z_1 & + & 5\, z_2 & + & 5\, z_3 & + & 5\, z_4 & \geq & 13 \\ & z_1, & & z_2, & & z_3, & & z_4 & \in & \{0,1\} \end{array}$$

  With $y_j = 1 - z_j$, the constant term and the right-hand side of the complementary problem are

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j = \frac{6}{5}, \qquad \sum_{j \in \mathscr{I}} w_j - (c+1) = 18 - 13 = 5,$$

  and the complementary problem $\text{ILP}^2_{\text{CSP}}$ is:

$$\begin{array}{lccccccccc} \max & & & & & & \frac{1}{5}\,y_3 & + & y_4 & \\ \text{subject to} & 3\,y_1 & + & 5\,y_2 & + & 5\,y_3 & + & 5\,y_4 & \leq & 5 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

with $z(\text{ILP}_{\text{CSP}}) = \frac{6}{5} - z(\text{ILP}^2_{\text{CSP}})$.

- **Greedy heuristic**
  The indices are scanned by non-increasing ratio q_j/w_j, starting from the residual right-hand side 5:

j	q_j	w_j	q_j/w_j	Decision	Residual
4	1	5	$\frac{1}{5}$	take	0
3	$\frac{1}{5}$	5	$\frac{1}{25}$	skip	0
1	0	3	0	$q_j = 0$	0
2	0	5	0	$q_j = 0$	0

  The greedy solution is $\bar{\boldsymbol{y}} = (0,0,0,1)$, with value 1. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{1,2,3\}$, of total coefficient $13 > 12$ and cost $\frac{6}{5} - 1 = \frac{1}{5} < 1$. Hence the cover inequality

$$x_1 + x_2 + x_3 \leq 2$$

  is violated by $\tilde{\boldsymbol{x}}$: $1 + 1 + \frac{4}{5} - 2 = \frac{4}{5}$.

- **Optimality check**
  Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{3,4\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 5 are: $\{3\}$ (value $\frac{1}{5}$), $\{4\}$ (value 1). Hence $z(\text{ILP}^2_{\text{CSP}}) = 1$, the greedy solution is optimal, $z(\text{ILP}_{\text{CSP}}) = \frac{1}{5} < 1$, and the maximum violation of a cover inequality is $\frac{4}{5}$.

5.2 Cover separation in three dimensions

Exercise 5.6 — Cover inequalities in 3D (*P1*)

Consider a binary problem with one constraint $\sum_j w_j x_j \leq c$, with positive integer coefficients w_j and right-hand side $c = 8$, on the binary variables indexed by the set $\mathscr{I} = \{1,2,3\}$ of $n = 3$ variables with the following objective coefficients and constraint coefficients:

j	1	2	3
p_j	12	4	2
w_j	5	4	3

The associated LP is:

$$
\begin{array}{rrcrcrcl}
\max & 12\,x_1 & + & 4\,x_2 & + & 2\,x_3 & & \\
\text{subject to} & 5\,x_1 & + & 4\,x_2 & + & 3\,x_3 & \leq & 8 \\
& x_1 & & & & & \leq & 1 \\
& & & x_2 & & & \leq & 1 \\
& & & & & x_3 & \leq & 1 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$

and the following points of its feasible region:

$$
\begin{aligned}
&(a) \quad \tilde{\boldsymbol{x}} = \left(1, \tfrac{3}{4}, 0\right), \\
&(b) \quad \tilde{\boldsymbol{x}} = \left(\tfrac{1}{5}, 1, 1\right), \\
&(c) \quad \tilde{\boldsymbol{x}} = \left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}\right).
\end{aligned}
$$

Questions For each point:

1. write the separation problem ILP_{CSP} and its complementary problem $\text{ILP}^2_{\text{CSP}}$;
2. apply the greedy heuristic to $\text{ILP}^2_{\text{CSP}}$ and, if possible, find a violated cover inequality;
3. check whether the greedy solution is optimal;
4. draw the LP relaxation together with the violated cover inequality, if any.

A subset of indices $C \subseteq \mathscr{I}$ with $\sum_{j \in C} w_j > c$ is a **cover**, and the linear inequality

$$\sum_{j \in C} x_j \leq |C| - 1$$

is the **cover inequality** associated with C. Given a point $\tilde{\boldsymbol{x}}$, a cover inequality violated by $\tilde{\boldsymbol{x}}$ exists if and only if $z(\text{ILP}_{\text{CSP}}) < 1$, where

$$(\text{ILP}_{\text{CSP}}) \qquad \min \left\{ \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j)\, z_j : \sum_{j \in \mathscr{I}} w_j\, z_j \geq c + 1,\ z_j \in \{0,1\}\ \forall j \in \mathscr{I} \right\}.$$

Solution

Separation of the cover inequalities violated by the three points.

- **Cover inequalities**
  All three points belong to the LP relaxation; the left-hand sides are

  $$5 + 4 \cdot \frac{3}{4} + 0 = 8, \qquad 5 \cdot \frac{1}{5} + 4 + 3 = 8, \qquad 5 \cdot \frac{1}{3} + 4 \cdot \frac{1}{3} + 3 \cdot \frac{1}{3} = 4 \leq 8.$$

  The covers are the subsets of indices whose total coefficient exceeds $c = 8$:

C	$\sum_{j\in C} w_j$	$\lvert C\rvert$	Cover inequality	Minimal
$\{1,2\}$	9	2	$x_1 + x_2 \le 1$	yes
$\{1,2,3\}$	12	3	$x_1 + x_2 + x_3 \le 2$	no

In all three cases, the right-hand side of the complementary problem is $\sum_{j\in\mathscr{I}} w_j - (c+1) = 12 - 9 = 3$.

- **Case (a): $\tilde{\boldsymbol{x}} = \left(1, \frac{3}{4}, 0\right)$**

a) **Separation problem**
The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(0, \frac{1}{4}, 1\right)$. The separation problem ILP_{CSP} (left) and, with $y_j = 1 - z_j$, the complementary problem $\text{ILP}^2_{\text{CSP}}$ (right) are:

$$\begin{array}{rrcrcrl} \min & & & \frac{1}{4}z_2 & + & z_3 & \\ \text{subject to} & 5z_1 & + & 4z_2 & + & 3z_3 & \ge 9 \\ & z_1, & & z_2, & & z_3 & \in \{0,1\} \end{array} \qquad \begin{array}{rrcrcrl} \max & & & \frac{1}{4}y_2 & + & y_3 & \\ \text{subject to} & 5y_1 & + & 4y_2 & + & 3y_3 & \le 3 \\ & y_1, & & y_2, & & y_3 & \in \{0,1\} \end{array}$$

with $z(\text{ILP}_{\text{CSP}}) = n - \sum_{j\in\mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{5}{4} - z(\text{ILP}^2_{\text{CSP}})$.

b) **Greedy heuristic**
The indices are scanned by non-increasing ratio q_j/w_j, starting from the residual right-hand side 3:

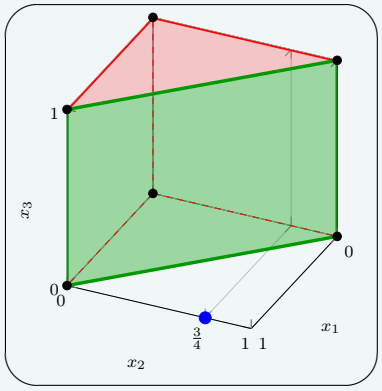
j	q_j	w_j	q_j/w_j	Decision	Residual
3	1	3	$\frac{1}{3}$	take	0
2	$\frac{1}{4}$	4	$\frac{1}{16}$	skip	0
1	0	5	0	$q_j = 0$	0

The greedy solution is $\bar{\boldsymbol{y}} = (0,0,1)$, with value 1. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{1,2\}$, of total coefficient $9 > 8$ and cost $\frac{5}{4} - 1 = \frac{1}{4} < 1$. Hence the cover inequality

$$x_1 + x_2 \le 1$$

is violated by $\tilde{\boldsymbol{x}}$: $1 + \frac{3}{4} - 1 = \frac{3}{4}$.

c) **Optimality check**
Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{2,3\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 3 are: $\{3\}$ (value 1). Hence $z(\text{ILP}^2_{\text{CSP}}) = 1$, the greedy solution is optimal, $z(\text{ILP}_{\text{CSP}}) = \frac{1}{4} < 1$, and the maximum violation of a cover inequality is $\frac{3}{4}$.



Case (a): the LP relaxation and the cover inequality $x_1 + x_2 \leq 1$.

- **Case (b): $\tilde{\boldsymbol{x}} = \left(\frac{1}{5}, 1, 1\right)$**

  a) **Separation problem**
  The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(\frac{4}{5}, 0, 0\right)$. The separation problem ILP_{CSP} (left) and, with $y_j = 1 - z_j$, the complementary problem $\text{ILP}^2_{\text{CSP}}$ (right) are:

$$\begin{aligned} \min \quad & \tfrac{4}{5} z_1 \\ \text{subject to} \quad & 5 z_1 + 4 z_2 + 3 z_3 \geq 9 \\ & z_1, \quad z_2, \quad z_3 \in \{0,1\} \end{aligned} \qquad \begin{aligned} \max \quad & \tfrac{4}{5} y_1 \\ \text{subject to} \quad & 5 y_1 + 4 y_2 + 3 y_3 \leq 3 \\ & y_1, \quad y_2, \quad y_3 \in \{0,1\} \end{aligned}$$

  with $z(\text{ILP}_{\text{CSP}}) = n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{4}{5} - z(\text{ILP}^2_{\text{CSP}})$.

  b) **Greedy heuristic**
  The indices are scanned by non-increasing ratio q_j / w_j, starting from the residual right-hand side 3:

j	q_j	w_j	q_j/w_j	Decision	Residual
1	$\frac{4}{5}$	5	$\frac{4}{25}$	skip	3
2	0	4	0	$q_j = 0$	3
3	0	3	0	$q_j = 0$	3

  The greedy solution is $\bar{\boldsymbol{y}} = (0,0,0)$, with value 0. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{1,2,3\}$, of total coefficient $12 > 8$ and cost $\frac{4}{5} - 0 = \frac{4}{5} < 1$. Hence the cover inequality

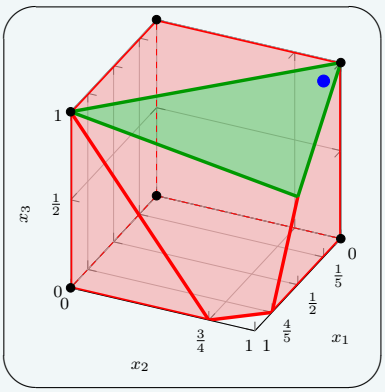
$$x_1 + x_2 + x_3 \leq 2$$

  is violated by $\tilde{\boldsymbol{x}}$: $\frac{1}{5} + 1 + 1 - 2 = \frac{1}{5}$.

  c) **Optimality check**
  No index with $q_j > 0$ satisfies the constraint with right-hand side 3 alone, hence every feasible solution of $\text{ILP}^2_{\text{CSP}}$ has value 0. Hence $z(\text{ILP}^2_{\text{CSP}}) = 0$, the greedy

solution is optimal, $z(\text{ILP}_{\text{CSP}}) = \frac{4}{5} < 1$, and the maximum violation of a cover inequality is $\frac{1}{5}$.

The cover $\bar{C}$ is not minimal: removing an index with $q_j = 0$ gives the minimal cover $\{1, 2\}$, with the same cost $\frac{4}{5}$. Its cover inequality $x_1 + x_2 \leq 1$ has the same violation $\frac{1}{5}$ and is stronger.



Case (b): the LP relaxation and the cover inequality $x_1 + x_2 + x_3 \leq 2$.

- **Case (c):** $\tilde{\boldsymbol{x}} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$

a) **Separation problem**

The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$. The separation problem ILP_{CSP} (left) and, with $y_j = 1 - z_j$, the complementary problem $\text{ILP}^2_{\text{CSP}}$ (right) are:

$$\begin{aligned}
\min \quad & \tfrac{2}{3} z_1 + \tfrac{2}{3} z_2 + \tfrac{2}{3} z_3 & & \\
\text{subject to} \quad & 5 z_1 + 4 z_2 + 3 z_3 & \geq & \ 9 \\
& z_1, \quad z_2, \quad z_3 & \in & \ \{0, 1\} \\
\max \quad & \tfrac{2}{3} y_1 + \tfrac{2}{3} y_2 + \tfrac{2}{3} y_3 & & \\
\text{subject to} \quad & 5 y_1 + 4 y_2 + 3 y_3 & \leq & \ 3 \\
& y_1, \quad y_2, \quad y_3 & \in & \ \{0, 1\}
\end{aligned}$$

with $z(\text{ILP}_{\text{CSP}}) = n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = 2 - z(\text{ILP}^2_{\text{CSP}})$.

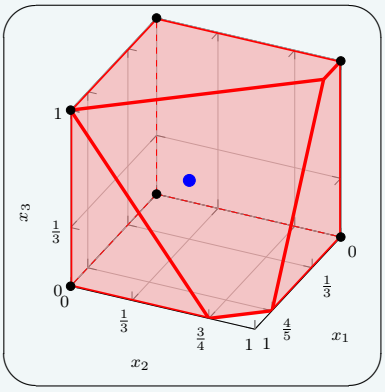
b) **Greedy heuristic**

The indices are scanned by non-increasing ratio q_j / w_j, starting from the residual right-hand side 3:

j	q_j	w_j	q_j/w_j	Decision	Residual
3	$\frac{2}{3}$	3	$\frac{2}{9}$	take	0
2	$\frac{2}{3}$	4	$\frac{1}{6}$	skip	0
1	$\frac{2}{3}$	5	$\frac{2}{15}$	skip	0

The greedy solution is $\bar{\boldsymbol{y}} = (0, 0, 1)$, with value $\frac{2}{3}$. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{1, 2\}$, with cost $2 - \frac{2}{3} = \frac{4}{3} \geq 1$: its cover inequality is not violated by $\tilde{\boldsymbol{x}}$.

c) **Optimality check**
Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{1, 2, 3\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 3 are: $\{3\}$ (value $\frac{2}{3}$). Hence $z(\text{ILP}^2_{\text{CSP}}) = \frac{2}{3}$, the greedy solution is optimal, and $z(\text{ILP}_{\text{CSP}}) = \frac{4}{3} \geq 1$: **no** cover inequality is violated by $\tilde{\boldsymbol{x}}$.



Case (c): the LP relaxation; no cover inequality is violated by the point.

Exercise 5.7 — Cover inequalities in 3D ($P2$)

Consider a binary problem with one constraint $\sum_j w_j x_j \leq c$, with positive integer coefficients w_j and right-hand side $c = 11$, on the binary variables indexed by the set $\mathscr{I} = \{1, 2, 3\}$ of $n = 3$ variables with the following objective coefficients and constraint coefficients:

j	1	2	3
p_j	20	11	4
w_j	2	3	9

The associated LP is:

$$
\begin{array}{rrcrcrcl}
\max & 20\,x_1 & + & 11\,x_2 & + & 4\,x_3 & & \\
\text{subject to} & 2\,x_1 & + & 3\,x_2 & + & 9\,x_3 & \leq & 11 \\
& x_1 & & & & & \leq & 1 \\
& & & x_2 & & & \leq & 1 \\
& & & & & x_3 & \leq & 1 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$

and the following points of its feasible region:

$$
\begin{aligned}
&(a) \quad \tilde{\boldsymbol{x}} = \left(1, 1, \tfrac{2}{3}\right), \\
&(b) \quad \tilde{\boldsymbol{x}} = \left(0, \tfrac{2}{3}, 1\right), \\
&(c) \quad \tilde{\boldsymbol{x}} = \left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}\right).
\end{aligned}
$$

Questions For each point:

1. write the separation problem ILP_{CSP} and its complementary problem $\text{ILP}^2_{\text{CSP}}$;
2. apply the greedy heuristic to $\text{ILP}^2_{\text{CSP}}$ and, if possible, find a violated cover inequality;
3. check whether the greedy solution is optimal;
4. draw the LP relaxation together with the violated cover inequality, if any.

A subset of indices $C \subseteq \mathscr{I}$ with $\sum_{j \in C} w_j > c$ is a **cover**, and the linear inequality

$$\sum_{j \in C} x_j \leq |C| - 1$$

is the **cover inequality** associated with C. Given a point $\tilde{\boldsymbol{x}}$, a cover inequality violated by $\tilde{\boldsymbol{x}}$ exists if and only if $z(\text{ILP}_{\text{CSP}}) < 1$, where

$$(\text{ILP}_{\text{CSP}}) \qquad \min \Big\{ \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j)\, z_j : \sum_{j \in \mathscr{I}} w_j\, z_j \geq c + 1,\ z_j \in \{0,1\}\ \forall j \in \mathscr{I} \Big\}.$$

Solution

Separation of the cover inequalities violated by the three points.

- **Cover inequalities**
  All three points belong to the LP relaxation; the left-hand sides are

  $$2 + 3 + 9 \cdot \frac{2}{3} = 11, \qquad 0 + 3 \cdot \frac{2}{3} + 9 = 11, \qquad 2 \cdot \frac{1}{3} + 3 \cdot \frac{1}{3} + 9 \cdot \frac{1}{3} = \frac{14}{3} \leq 11.$$

  The covers are the subsets of indices whose total coefficient exceeds $c = 11$:

C	$\sum_{j \in C} w_j$	$\lvert C \rvert$	Cover inequality	Minimal
$\{2,3\}$	12	2	$x_2 + x_3 \leq 1$	yes
$\{1,2,3\}$	14	3	$x_1 + x_2 + x_3 \leq 2$	no

  In all three cases, the right-hand side of the complementary problem is $\sum_{j \in \mathscr{I}} w_j - (c+1) = 14 - 12 = 2$.

- **Case (a): $\tilde{\boldsymbol{x}} = \left(1, 1, \frac{2}{3}\right)$**

  a) **Separation problem**
  The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(0, 0, \frac{1}{3}\right)$. The separation problem ILP_{CSP} (left) and, with $y_j = 1 - z_j$, the complementary problem $\text{ILP}^2_{\text{CSP}}$ (right) are:

$$\begin{array}{lrcrcrcl} \min & & & & & \frac{1}{3}\, z_3 & & \\ \text{subject to} & 2\, z_1 & + & 3\, z_2 & + & 9\, z_3 & \geq & 12 \\ & z_1, & & z_2, & & z_3 & \in & \{0,1\} \end{array}$$

$$\begin{array}{lrcrcrcl} \max & & & & & \frac{1}{3}\,y_3 & & \\ \text{subject to} & 2\,y_1 & + & 3\,y_2 & + & 9\,y_3 & \leq & 2 \\ & y_1, & & y_2, & & y_3 & \in & \{0,1\} \end{array}$$

with $z(\text{ILP}_{\text{CSP}}) = n - \sum_{j\in\mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{1}{3} - z(\text{ILP}^2_{\text{CSP}})$.

b) **Greedy heuristic**

The indices are scanned by non-increasing ratio q_j/w_j, starting from the residual right-hand side 2:

j	q_j	w_j	q_j/w_j	Decision	Residual
3	$\frac{1}{3}$	9	$\frac{1}{27}$	skip	2
1	0	2	0	$q_j = 0$	2
2	0	3	0	$q_j = 0$	2

The greedy solution is $\bar{\boldsymbol{y}} = (0,0,0)$, with value 0. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{1,2,3\}$, of total coefficient $14 > 11$ and cost $\frac{1}{3} - 0 = \frac{1}{3} < 1$. Hence the cover inequality

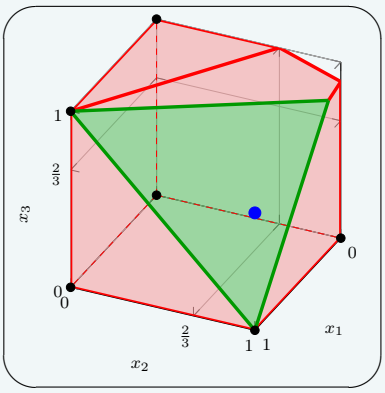
$$x_1 + x_2 + x_3 \leq 2$$

is violated by $\tilde{\boldsymbol{x}}$: $1 + 1 + \frac{2}{3} - 2 = \frac{2}{3}$.

c) **Optimality check**

No index with $q_j > 0$ satisfies the constraint with right-hand side 2 alone, hence every feasible solution of $\text{ILP}^2_{\text{CSP}}$ has value 0. Hence $z(\text{ILP}^2_{\text{CSP}}) = 0$, the greedy solution is optimal, $z(\text{ILP}_{\text{CSP}}) = \frac{1}{3} < 1$, and the maximum violation of a cover inequality is $\frac{2}{3}$.

The cover $\bar{C}$ is not minimal: removing an index with $q_j = 0$ gives the minimal cover $\{2,3\}$, with the same cost $\frac{1}{3}$. Its cover inequality $x_2 + x_3 \leq 1$ has the same violation $\frac{2}{3}$ and is stronger.



Case (a): the LP relaxation and the cover inequality $x_1 + x_2 + x_3 \leq 2$.

- **Case (b):** $\tilde{\boldsymbol{x}} = \left(0, \frac{2}{3}, 1\right)$

a) **Separation problem**
The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(1, \frac{1}{3}, 0\right)$. The separation problem ILP_{CSP} (left) and, with $y_j = 1 - z_j$, the complementary problem $\text{ILP}^2_{\text{CSP}}$ (right) are:

$$\begin{array}{rrcrcrcl} \min & z_1 & + & \frac{1}{3} z_2 & & & & \\ \text{subject to} & 2\, z_1 & + & 3\, z_2 & + & 9\, z_3 & \geq & 12 \\ & z_1, & & z_2, & & z_3 & \in & \{0,1\} \\ \max & y_1 & + & \frac{1}{3} y_2 & & & & \\ \text{subject to} & 2\, y_1 & + & 3\, y_2 & + & 9\, y_3 & \leq & 2 \\ & y_1, & & y_2, & & y_3 & \in & \{0,1\} \end{array}$$

with $z(\text{ILP}_{\text{CSP}}) = n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{4}{3} - z(\text{ILP}^2_{\text{CSP}})$.

b) **Greedy heuristic**
The indices are scanned by non-increasing ratio q_j / w_j, starting from the residual right-hand side 2:

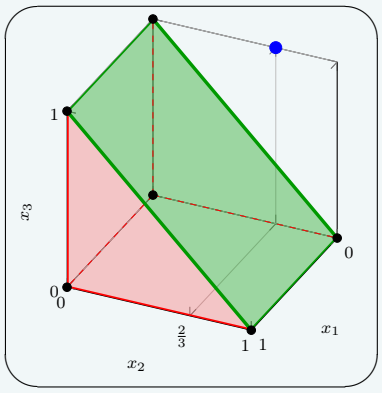
j	q_j	w_j	q_j / w_j	Decision	Residual
1	1	2	$\frac{1}{2}$	take	0
2	$\frac{1}{3}$	3	$\frac{1}{9}$	skip	0
3	0	9	0	$q_j = 0$	0

The greedy solution is $\bar{\boldsymbol{y}} = (1, 0, 0)$, with value 1. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{2, 3\}$, of total coefficient $12 > 11$ and cost $\frac{4}{3} - 1 = \frac{1}{3} < 1$. Hence the cover inequality

$$x_2 + x_3 \leq 1$$

is violated by $\tilde{\boldsymbol{x}}$: $\frac{2}{3} + 1 - 1 = \frac{2}{3}$.

c) **Optimality check**
Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{1, 2\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 2 are: $\{1\}$ (value 1). Hence $z(\text{ILP}^2_{\text{CSP}}) = 1$, the greedy solution is optimal, $z(\text{ILP}_{\text{CSP}}) = \frac{1}{3} < 1$, and the maximum violation of a cover inequality is $\frac{2}{3}$.



Case (b): the LP relaxation and the cover inequality $x_2 + x_3 \leq 1$.

- **Case (c):** $\tilde{\boldsymbol{x}} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$

a) **Separation problem**
The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$. The separation problem ILP_{CSP} (left) and, with $y_j = 1 - z_j$, the complementary problem $\text{ILP}^2_{\text{CSP}}$ (right) are:

$$\begin{array}{rrcrcrcl} \min & \frac{2}{3}z_1 & + & \frac{2}{3}z_2 & + & \frac{2}{3}z_3 & & \\ \text{subject to} & 2z_1 & + & 3z_2 & + & 9z_3 & \geq & 12 \\ & z_1, & & z_2, & & z_3 & \in & \{0,1\} \\ \max & \frac{2}{3}y_1 & + & \frac{2}{3}y_2 & + & \frac{2}{3}y_3 & & \\ \text{subject to} & 2y_1 & + & 3y_2 & + & 9y_3 & \leq & 2 \\ & y_1, & & y_2, & & y_3 & \in & \{0,1\} \end{array}$$

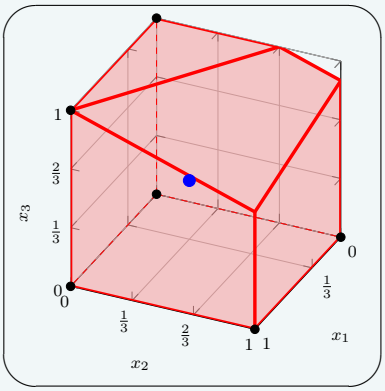
with $z(\text{ILP}_{\text{CSP}}) = n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = 2 - z(\text{ILP}^2_{\text{CSP}})$.

b) **Greedy heuristic**
The indices are scanned by non-increasing ratio q_j/w_j, starting from the residual right-hand side 2:

j	q_j	w_j	q_j/w_j	Decision	Residual
1	$\frac{2}{3}$	2	$\frac{1}{3}$	take	0
2	$\frac{2}{3}$	3	$\frac{2}{9}$	skip	0
3	$\frac{2}{3}$	9	$\frac{2}{27}$	skip	0

The greedy solution is $\bar{\boldsymbol{y}} = (1, 0, 0)$, with value $\frac{2}{3}$. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{2, 3\}$, with cost $2 - \frac{2}{3} = \frac{4}{3} \geq 1$: its cover inequality is not violated by $\tilde{\boldsymbol{x}}$.

c) **Optimality check**
Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{1, 2, 3\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 2 are: $\{1\}$ (value $\frac{2}{3}$). Hence $z(\text{ILP}^2_{\text{CSP}}) = \frac{2}{3}$, the greedy solution is optimal, and $z(\text{ILP}_{\text{CSP}}) = \frac{4}{3} \geq 1$: **no** cover inequality is violated by $\tilde{\boldsymbol{x}}$.



Case (c): the LP relaxation; no cover inequality is violated by the point.

Exercise 5.8 — Cover inequalities in 3D ($P3$)

Consider a binary problem with one constraint $\sum_j w_j x_j \leq c$, with positive integer coefficients w_j and right-hand side $c = 9$, on the binary variables indexed by the set $\mathscr{I} = \{1, 2, 3\}$ of $n = 3$ variables with the following objective coefficients and constraint coefficients:

j	1	2	3
p_j	16	19	8
w_j	5	6	7

The associated LP is:

$$\begin{array}{rrcrcrcl} \max & 16\,x_1 & + & 19\,x_2 & + & 8\,x_3 & & \\ \text{subject to} & 5\,x_1 & + & 6\,x_2 & + & 7\,x_3 & \leq & 9 \\ & x_1 & & & & & \leq & 1 \\ & & & x_2 & & & \leq & 1 \\ & & & & & x_3 & \leq & 1 \\ & x_1, & & x_2, & & x_3 & \geq & 0 \end{array}$$

and the following points of its feasible region:

$$\begin{aligned} &(a) \quad \tilde{\boldsymbol{x}} = \left(1, \tfrac{2}{3}, 0\right), \\ &(b) \quad \tilde{\boldsymbol{x}} = \left(0, \tfrac{1}{3}, 1\right), \\ &(c) \quad \tilde{\boldsymbol{x}} = \left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}\right). \end{aligned}$$

Questions For each point:

1. write the separation problem ILP_{CSP} and its complementary problem $\text{ILP}^2_{\text{CSP}}$;
2. apply the greedy heuristic to $\text{ILP}^2_{\text{CSP}}$ and, if possible, find a violated cover inequality;
3. check whether the greedy solution is optimal;
4. draw the LP relaxation together with the violated cover inequality, if any.

A subset of indices $C \subseteq \mathscr{I}$ with $\sum_{j \in C} w_j > c$ is a **cover**, and the linear inequality

$$\sum_{j \in C} x_j \leq |C| - 1$$

is the **cover inequality** associated with C. Given a point $\tilde{\boldsymbol{x}}$, a cover inequality violated by $\tilde{\boldsymbol{x}}$ exists if and only if $z(\text{ILP}_{\text{CSP}}) < 1$, where

$$(\text{ILP}_{\text{CSP}}) \qquad \min \left\{ \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j)\, z_j : \sum_{j \in \mathscr{I}} w_j\, z_j \geq c + 1,\ z_j \in \{0, 1\}\ \forall j \in \mathscr{I} \right\}.$$

Solution

Separation of the cover inequalities violated by the three points.

- **Cover inequalities**
  All three points belong to the LP relaxation; the left-hand sides are

$$5+6\cdot\frac{2}{3}+0=9, \qquad 0+6\cdot\frac{1}{3}+7=9, \qquad 5\cdot\frac{1}{3}+6\cdot\frac{1}{3}+7\cdot\frac{1}{3}=6\le 9.$$

  The covers are the subsets of indices whose total coefficient exceeds $c=9$:

C	$\sum_{j\in C} w_j$	$\lvert C\rvert$	Cover inequality	Minimal
$\{1,2\}$	11	2	$x_1+x_2\le 1$	yes
$\{1,3\}$	12	2	$x_1+x_3\le 1$	yes
$\{2,3\}$	13	2	$x_2+x_3\le 1$	yes
$\{1,2,3\}$	18	3	$x_1+x_2+x_3\le 2$	no

  In all three cases, the right-hand side of the complementary problem is $\sum_{j\in\mathscr{J}} w_j-(c+1)=18-10=8$.

- **Case (a): $\tilde{\boldsymbol{x}}=\left(1,\frac{2}{3},0\right)$**

  a) **Separation problem**
  The costs $q_j=1-\tilde{x}_j$ are $q=\left(0,\frac{1}{3},1\right)$. The separation problem $\mathrm{ILP}_{\mathrm{CSP}}$ (left) and, with $y_j=1-z_j$, the complementary problem $\mathrm{ILP}^2_{\mathrm{CSP}}$ (right) are:

$$\begin{array}{rrcrcrl} \min & & & \frac{1}{3}z_2 & + & z_3 & \\ \text{subject to} & 5z_1 & + & 6z_2 & + & 7z_3 & \ge\ 10 \\ & z_1, & & z_2, & & z_3 & \in\ \{0,1\} \\ \max & & & \frac{1}{3}y_2 & + & y_3 & \\ \text{subject to} & 5y_1 & + & 6y_2 & + & 7y_3 & \le\ 8 \\ & y_1, & & y_2, & & y_3 & \in\ \{0,1\} \end{array}$$

  with $z(\mathrm{ILP}_{\mathrm{CSP}})=n-\sum_{j\in\mathscr{J}}\tilde{x}_j-z(\mathrm{ILP}^2_{\mathrm{CSP}})=\frac{4}{3}-z(\mathrm{ILP}^2_{\mathrm{CSP}})$.

  b) **Greedy heuristic**
  The indices are scanned by non-increasing ratio q_j/w_j, starting from the residual right-hand side 8:

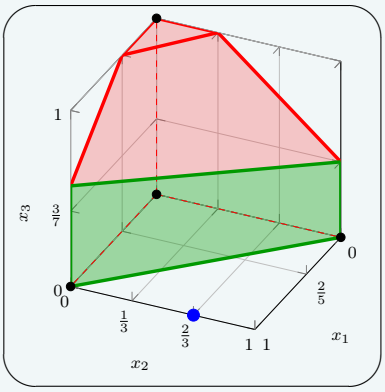
j	q_j	w_j	q_j/w_j	Decision	Residual
3	1	7	$\frac{1}{7}$	take	1
2	$\frac{1}{3}$	6	$\frac{1}{18}$	skip	1
1	0	5	0	$q_j=0$	1

  The greedy solution is $\bar{\boldsymbol{y}}=(0,0,1)$, with value 1. The corresponding cover is $\bar{C}=\{j\in\mathscr{J}:\bar{y}_j=0\}=\{1,2\}$, of total coefficient $11>9$ and cost $\frac{4}{3}-1=\frac{1}{3}<1$. Hence the cover inequality

$$x_1+x_2\le 1$$

  is violated by $\tilde{\boldsymbol{x}}$: $1+\frac{2}{3}-1=\frac{2}{3}$.

c) **Optimality check**
Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{2, 3\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 8 are: $\{2\}$ (value $\frac{1}{3}$), $\{3\}$ (value 1). Hence $z(\text{ILP}^2_{\text{CSP}}) = 1$, the greedy solution is optimal, $z(\text{ILP}_{\text{CSP}}) = \frac{1}{3} < 1$, and the maximum violation of a cover inequality is $\frac{2}{3}$.



Case (a): the LP relaxation and the cover inequality $x_1 + x_2 \leq 1$.

- **Case (b): $\tilde{\boldsymbol{x}} = \left(0, \frac{1}{3}, 1\right)$**

a) **Separation problem**
The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(1, \frac{2}{3}, 0\right)$. The separation problem ILP_{CSP} (left) and, with $y_j = 1 - z_j$, the complementary problem $\text{ILP}^2_{\text{CSP}}$ (right) are:

$$\begin{array}{rrcrcrcl} \min & z_1 & + & \frac{2}{3}\, z_2 & & & & \\ \text{subject to} & 5\, z_1 & + & 6\, z_2 & + & 7\, z_3 & \geq & 10 \\ & z_1, & & z_2, & & z_3 & \in & \{0, 1\} \\ \max & y_1 & + & \frac{2}{3}\, y_2 & & & & \\ \text{subject to} & 5\, y_1 & + & 6\, y_2 & + & 7\, y_3 & \leq & 8 \\ & y_1, & & y_2, & & y_3 & \in & \{0, 1\} \end{array}$$

with $z(\text{ILP}_{\text{CSP}}) = n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{5}{3} - z(\text{ILP}^2_{\text{CSP}})$.

b) **Greedy heuristic**
The indices are scanned by non-increasing ratio q_j / w_j, starting from the residual right-hand side 8:

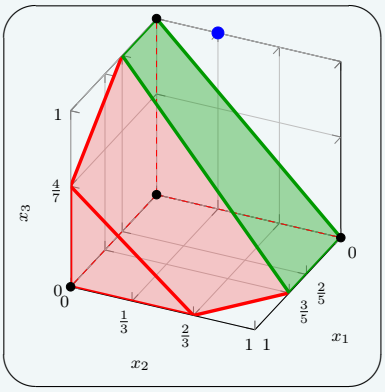
j	q_j	w_j	q_j / w_j	Decision	Residual
1	1	5	$\frac{1}{5}$	take	3
2	$\frac{2}{3}$	6	$\frac{1}{9}$	skip	3
3	0	7	0	$q_j = 0$	3

The greedy solution is $\bar{\boldsymbol{y}} = (1, 0, 0)$, with value 1. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{2, 3\}$, of total coefficient $13 > 9$ and cost $\frac{5}{3} - 1 = \frac{2}{3} < 1$. Hence the cover inequality

$$x_2 + x_3 \leq 1$$

is violated by $\tilde{\boldsymbol{x}}$: $\frac{1}{3} + 1 - 1 = \frac{1}{3}$.

c) **Optimality check**
Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{1, 2\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 8 are: $\{1\}$ (value 1), $\{2\}$ (value $\frac{2}{3}$). Hence $z(\text{ILP}^2_{\text{CSP}}) = 1$, the greedy solution is optimal, $z(\text{ILP}_{\text{CSP}}) = \frac{2}{3} < 1$, and the maximum violation of a cover inequality is $\frac{1}{3}$.



Case (b): the LP relaxation and the cover inequality $x_2 + x_3 \leq 1$.

- **Case (c): $\tilde{\boldsymbol{x}} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$**

a) **Separation problem**
The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$. The separation problem ILP_{CSP} (left) and, with $y_j = 1 - z_j$, the complementary problem $\text{ILP}^2_{\text{CSP}}$ (right) are:

$$\begin{array}{rrrrrrrrl} \min & \frac{2}{3}z_1 & + & \frac{2}{3}z_2 & + & \frac{2}{3}z_3 & & & \\ \text{subject to} & 5z_1 & + & 6z_2 & + & 7z_3 & \geq & 10 & \\ & z_1, & & z_2, & & z_3 & \in & \{0,1\} & \\ \max & \frac{2}{3}y_1 & + & \frac{2}{3}y_2 & + & \frac{2}{3}y_3 & & & \\ \text{subject to} & 5y_1 & + & 6y_2 & + & 7y_3 & \leq & 8 & \\ & y_1, & & y_2, & & y_3 & \in & \{0,1\} & \end{array}$$

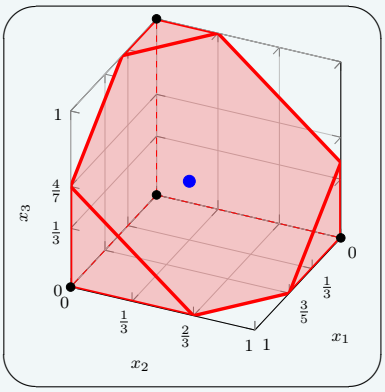
with $z(\text{ILP}_{\text{CSP}}) = n - \sum_{j \in \mathscr{J}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = 2 - z(\text{ILP}^2_{\text{CSP}})$.

b) **Greedy heuristic**
The indices are scanned by non-increasing ratio q_j/w_j, starting from the residual right-hand side 8:

j	q_j	w_j	q_j/w_j	Decision	Residual
1	$\frac{2}{3}$	5	$\frac{2}{15}$	take	3
2	$\frac{2}{3}$	6	$\frac{1}{9}$	skip	3
3	$\frac{2}{3}$	7	$\frac{2}{21}$	skip	3

The greedy solution is $\bar{\boldsymbol{y}} = (1, 0, 0)$, with value $\frac{2}{3}$. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{2, 3\}$, with cost $2 - \frac{2}{3} = \frac{4}{3} \geq 1$: its cover inequality is not violated by $\tilde{\boldsymbol{x}}$.

c) **Optimality check**
Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{1, 2, 3\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 8 are: $\{1\}$ (value $\frac{2}{3}$), $\{2\}$ (value $\frac{2}{3}$), $\{3\}$ (value $\frac{2}{3}$). Hence $z(\text{ILP}^2_{\text{CSP}}) = \frac{2}{3}$, the greedy solution is optimal, and $z(\text{ILP}_{\text{CSP}}) = \frac{4}{3} \geq 1$: **no** cover inequality is violated by $\tilde{\boldsymbol{x}}$.



Case (c): the LP relaxation; no cover inequality is violated by the point.

Exercise 5.9 — Cover inequalities in 3D ($P4$)

Consider a binary problem with one constraint $\sum_j w_j x_j \leq c$, with positive integer coefficients w_j and right-hand side $c = 5$, on the binary variables indexed by the set $\mathscr{I} = \{1, 2, 3\}$ of $n = 3$ variables with the following objective coefficients and constraint coefficients:

j	1	2	3
p_j	14	17	8
w_j	2	5	5

The associated LP is:

$$
\begin{array}{rrcrcrcl}
\max & 14\,x_1 & + & 17\,x_2 & + & 8\,x_3 & & \\
\text{subject to} & 2\,x_1 & + & 5\,x_2 & + & 5\,x_3 & \leq & 5 \\
& x_1 & & & & & \leq & 1 \\
& & & x_2 & & & \leq & 1 \\
& & & & & x_3 & \leq & 1 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$

and the following points of its feasible region:

$$(a) \quad \tilde{\boldsymbol{x}} = \left(1, \tfrac{3}{5}, 0\right),$$
$$(b) \quad \tilde{\boldsymbol{x}} = \left(1, 0, \tfrac{3}{5}\right),$$
$$(c) \quad \tilde{\boldsymbol{x}} = \left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}\right).$$

Questions For each point:

1. write the separation problem ILP_{CSP} and its complementary problem $\text{ILP}^2_{\text{CSP}}$;
2. apply the greedy heuristic to $\text{ILP}^2_{\text{CSP}}$ and, if possible, find a violated cover inequality;
3. check whether the greedy solution is optimal;
4. draw the LP relaxation together with the violated cover inequality, if any.

A subset of indices $C \subseteq \mathscr{I}$ with $\sum_{j \in C} w_j > c$ is a **cover**, and the linear inequality

$$\sum_{j \in C} x_j \leq |C| - 1$$

is the **cover inequality** associated with C. Given a point $\tilde{\boldsymbol{x}}$, a cover inequality violated by $\tilde{\boldsymbol{x}}$ exists if and only if $z(\text{ILP}_{\text{CSP}}) < 1$, where

$$(\text{ILP}_{\text{CSP}}) \qquad \min \left\{ \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j)\, z_j : \sum_{j \in \mathscr{I}} w_j\, z_j \geq c + 1,\ z_j \in \{0,1\}\ \forall j \in \mathscr{I} \right\}.$$

Solution

Separation of the cover inequalities violated by the three points.

- **Cover inequalities**
  All three points belong to the LP relaxation; the left-hand sides are

$$2 + 5 \cdot \frac{3}{5} + 0 = 5, \qquad 2 + 0 + 5 \cdot \frac{3}{5} = 5, \qquad 2 \cdot \frac{1}{3} + 5 \cdot \frac{1}{3} + 5 \cdot \frac{1}{3} = 4 \leq 5.$$

  The covers are the subsets of indices whose total coefficient exceeds $c = 5$:

C	$\sum_{j \in C} w_j$	$\lvert C \rvert$	Cover inequality	Minimal
$\{1,2\}$	7	2	$x_1 + x_2 \leq 1$	yes
$\{1,3\}$	7	2	$x_1 + x_3 \leq 1$	yes
$\{2,3\}$	10	2	$x_2 + x_3 \leq 1$	yes
$\{1,2,3\}$	12	3	$x_1 + x_2 + x_3 \leq 2$	no

  In all three cases, the right-hand side of the complementary problem is $\sum_{j \in \mathscr{I}} w_j - (c+1) = 12 - 6 = 6$.

- **Case (a):** $\tilde{\boldsymbol{x}} = \left(1, \frac{3}{5}, 0\right)$

a) **Separation problem**
The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(0, \frac{2}{5}, 1\right)$. The separation problem ILP_{CSP} (left) and, with $y_j = 1 - z_j$, the complementary problem $\text{ILP}^2_{\text{CSP}}$ (right) are:

$$\begin{array}{rrcrcrcl} \min & & & \frac{2}{5}\,z_2 & + & z_3 & & \\ \text{subject to} & 2\,z_1 & + & 5\,z_2 & + & 5\,z_3 & \geq & 6 \\ & z_1, & & z_2, & & z_3 & \in & \{0,1\} \\ \max & & & \frac{2}{5}\,y_2 & + & y_3 & & \\ \text{subject to} & 2\,y_1 & + & 5\,y_2 & + & 5\,y_3 & \leq & 6 \\ & y_1, & & y_2, & & y_3 & \in & \{0,1\} \end{array}$$

with $z(\text{ILP}_{\text{CSP}}) = n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{7}{5} - z(\text{ILP}^2_{\text{CSP}})$.

b) **Greedy heuristic**
The indices are scanned by non-increasing ratio q_j/w_j, starting from the residual right-hand side 6:

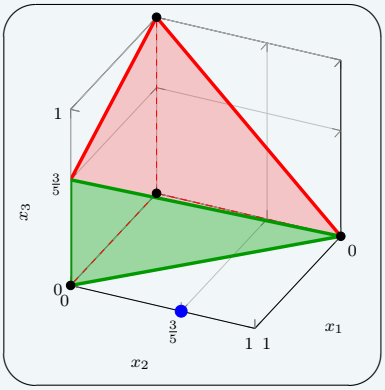
j	q_j	w_j	q_j/w_j	Decision	Residual
3	1	5	$\frac{1}{5}$	take	1
2	$\frac{2}{5}$	5	$\frac{2}{25}$	skip	1
1	0	2	0	$q_j = 0$	1

The greedy solution is $\bar{\boldsymbol{y}} = (0, 0, 1)$, with value 1. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{1, 2\}$, of total coefficient $7 > 5$ and cost $\frac{7}{5} - 1 = \frac{2}{5} < 1$. Hence the cover inequality

$$x_1 + x_2 \leq 1$$

is violated by $\tilde{\boldsymbol{x}}$: $1 + \frac{3}{5} - 1 = \frac{3}{5}$.

c) **Optimality check**
Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{2, 3\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 6 are: $\{2\}$ (value $\frac{2}{5}$), $\{3\}$ (value 1). Hence $z(\text{ILP}^2_{\text{CSP}}) = 1$, the greedy solution is optimal, $z(\text{ILP}_{\text{CSP}}) = \frac{2}{5} < 1$, and the maximum violation of a cover inequality is $\frac{3}{5}$.



Case (a): the LP relaxation and the cover inequality $x_1 + x_2 \leq 1$.

- **Case (b): $\tilde{\boldsymbol{x}} = \left(1, 0, \frac{3}{5}\right)$**

  a) **Separation problem**
  The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(0, 1, \frac{2}{5}\right)$. The separation problem ILP_{CSP} (left) and, with $y_j = 1 - z_j$, the complementary problem $\text{ILP}^2_{\text{CSP}}$ (right) are:

$$\begin{array}{rlllll} \min & & z_2 & + \frac{2}{5} z_3 & & \\ \text{subject to} & 2 z_1 + & 5 z_2 & + 5 z_3 & \geq & 6 \\ & z_1, & z_2, & z_3 & \in & \{0,1\} \end{array}$$

$$\begin{array}{rlllll} \max & & y_2 & + \frac{2}{5} y_3 & & \\ \text{subject to} & 2 y_1 + & 5 y_2 & + 5 y_3 & \leq & 6 \\ & y_1, & y_2, & y_3 & \in & \{0,1\} \end{array}$$

  with $z(\text{ILP}_{\text{CSP}}) = n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{7}{5} - z(\text{ILP}^2_{\text{CSP}})$.

  b) **Greedy heuristic**
  The indices are scanned by non-increasing ratio q_j / w_j, starting from the residual right-hand side 6:

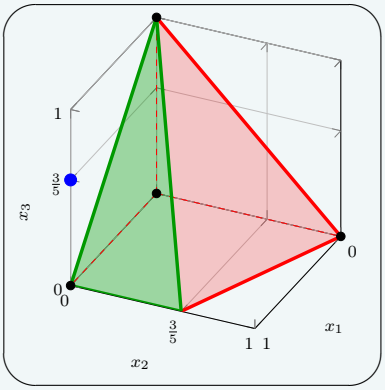
j	q_j	w_j	q_j/w_j	Decision	Residual
2	1	5	$\frac{1}{5}$	take	1
3	$\frac{2}{5}$	5	$\frac{2}{25}$	skip	1
1	0	2	0	$q_j = 0$	1

  The greedy solution is $\bar{\boldsymbol{y}} = (0, 1, 0)$, with value 1. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{1, 3\}$, of total coefficient $7 > 5$ and cost $\frac{7}{5} - 1 = \frac{2}{5} < 1$. Hence the cover inequality

$$x_1 + x_3 \leq 1$$

  is violated by $\tilde{\boldsymbol{x}}$: $1 + \frac{3}{5} - 1 = \frac{3}{5}$.

  c) **Optimality check**
  Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{2, 3\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 6 are: $\{2\}$ (value 1), $\{3\}$ (value $\frac{2}{5}$). Hence $z(\text{ILP}^2_{\text{CSP}}) = 1$, the greedy solution is optimal, $z(\text{ILP}_{\text{CSP}}) = \frac{2}{5} < 1$, and the maximum violation of a cover inequality is $\frac{3}{5}$.



Case (b): the LP relaxation and the cover inequality $x_1 + x_3 \leq 1$.

- **Case (c): $\tilde{\boldsymbol{x}} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$**

  a) **Separation problem**
  The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$. The separation problem ILP_{CSP} (left) and, with $y_j = 1 - z_j$, the complementary problem $\text{ILP}^2_{\text{CSP}}$ (right) are:

$$\begin{aligned}
\min \quad & \tfrac{2}{3} z_1 + \tfrac{2}{3} z_2 + \tfrac{2}{3} z_3 & \\
\text{subject to} \quad & 2 z_1 + 5 z_2 + 5 z_3 \geq 6 & \\
& z_1, \quad z_2, \quad z_3 \in \{0, 1\} &
\end{aligned}$$

$$\begin{aligned}
\max \quad & \tfrac{2}{3} y_1 + \tfrac{2}{3} y_2 + \tfrac{2}{3} y_3 & \\
\text{subject to} \quad & 2 y_1 + 5 y_2 + 5 y_3 \leq 6 & \\
& y_1, \quad y_2, \quad y_3 \in \{0, 1\} &
\end{aligned}$$

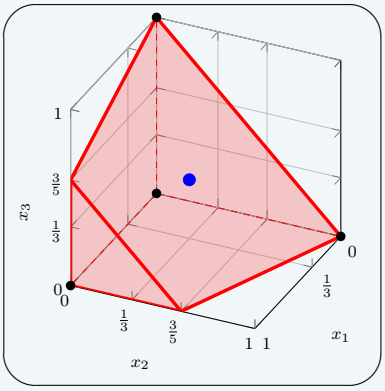
  with $z(\text{ILP}_{\text{CSP}}) = n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = 2 - z(\text{ILP}^2_{\text{CSP}})$.

  b) **Greedy heuristic**
  The indices are scanned by non-increasing ratio q_j / w_j, starting from the residual right-hand side 6:

j	q_j	w_j	q_j / w_j	Decision	Residual
1	$\frac{2}{3}$	2	$\frac{1}{3}$	take	4
2	$\frac{2}{3}$	5	$\frac{2}{15}$	skip	4
3	$\frac{2}{3}$	5	$\frac{2}{15}$	skip	4

  The greedy solution is $\bar{\boldsymbol{y}} = (1, 0, 0)$, with value $\frac{2}{3}$. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{2, 3\}$, with cost $2 - \frac{2}{3} = \frac{4}{3} \geq 1$: its cover inequality is not violated by $\tilde{\boldsymbol{x}}$.

  c) **Optimality check**
  Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{1, 2, 3\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 6 are: $\{1\}$ (value $\frac{2}{3}$), $\{2\}$ (value $\frac{2}{3}$), $\{3\}$ (value $\frac{2}{3}$). Hence $z(\text{ILP}^2_{\text{CSP}}) = \frac{2}{3}$, the greedy solution is optimal, and $z(\text{ILP}_{\text{CSP}}) = \frac{4}{3} \geq 1$: **no** cover inequality is violated by $\tilde{\boldsymbol{x}}$.



Case (c): the LP relaxation; no cover inequality is violated by the point.

Exercise 5.10 — Cover inequalities in 3D ($P5$)

Consider a binary problem with one constraint $\sum_j w_j x_j \leq c$, with positive integer coefficients w_j and right-hand side $c = 10$, on the binary variables indexed by the set $\mathscr{J} = \{1, 2, 3\}$ of $n = 3$ variables with the following objective coefficients and constraint coefficients:

j	1	2	3
p_j	13	10	3
w_j	5	4	6

The associated LP is:

$$
\begin{array}{rrcrcrcl}
\max & 13\,x_1 & + & 10\,x_2 & + & 3\,x_3 & & \\
\text{subject to} & 5\,x_1 & + & 4\,x_2 & + & 6\,x_3 & \leq & 10 \\
& x_1 & & & & & \leq & 1 \\
& & & x_2 & & & \leq & 1 \\
& & & & & x_3 & \leq & 1 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$

and the following points of its feasible region:

$$
\begin{aligned}
&(a) \quad \tilde{\boldsymbol{x}} = \left(1, 1, \tfrac{1}{6}\right), \\
&(b) \quad \tilde{\boldsymbol{x}} = \left(\tfrac{4}{5}, 0, 1\right), \\
&(c) \quad \tilde{\boldsymbol{x}} = \left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}\right).
\end{aligned}
$$

Questions For each point:

1. write the separation problem ILP_{CSP} and its complementary problem $\text{ILP}^2_{\text{CSP}}$;
2. apply the greedy heuristic to $\text{ILP}^2_{\text{CSP}}$ and, if possible, find a violated cover inequality;
3. check whether the greedy solution is optimal;
4. draw the LP relaxation together with the violated cover inequality, if any.

A subset of indices $C \subseteq \mathscr{I}$ with $\sum_{j \in C} w_j > c$ is a **cover**, and the linear inequality

$$\sum_{j \in C} x_j \leq |C| - 1$$

is the **cover inequality** associated with C. Given a point $\tilde{\boldsymbol{x}}$, a cover inequality violated by $\tilde{\boldsymbol{x}}$ exists if and only if $z(\text{ILP}_{\text{CSP}}) < 1$, where

$$(\text{ILP}_{\text{CSP}}) \qquad \min \Big\{ \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j)\, z_j : \sum_{j \in \mathscr{I}} w_j\, z_j \geq c + 1,\ z_j \in \{0,1\}\ \forall j \in \mathscr{I} \Big\}.$$

Solution

Separation of the cover inequalities violated by the three points.

- **Cover inequalities**
  All three points belong to the LP relaxation; the left-hand sides are

  $$5 + 4 + 6 \cdot \frac{1}{6} = 10, \qquad 5 \cdot \frac{4}{5} + 0 + 6 = 10, \qquad 5 \cdot \frac{1}{3} + 4 \cdot \frac{1}{3} + 6 \cdot \frac{1}{3} = 5 \leq 10.$$

  The covers are the subsets of indices whose total coefficient exceeds $c = 10$:

C	$\sum_{j \in C} w_j$	$\lvert C \rvert$	Cover inequality	Minimal
$\{1,3\}$	11	2	$x_1 + x_3 \leq 1$	yes
$\{1,2,3\}$	15	3	$x_1 + x_2 + x_3 \leq 2$	no

  In all three cases, the right-hand side of the complementary problem is $\sum_{j \in \mathscr{I}} w_j - (c+1) = 15 - 11 = 4$.

- **Case (a): $\tilde{\boldsymbol{x}} = \left(1, 1, \frac{1}{6}\right)$**

  a) **Separation problem**
  The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(0, 0, \frac{5}{6}\right)$. The separation problem ILP_{CSP} (left) and, with $y_j = 1 - z_j$, the complementary problem $\text{ILP}^2_{\text{CSP}}$ (right) are:

$$\begin{array}{rrrrrrrrr}
\min & & & & & \frac{5}{6}\, z_3 & & \\
\text{subject to} & 5\, z_1 & + & 4\, z_2 & + & 6\, z_3 & \geq & 11 \\
& z_1, & & z_2, & & z_3 & \in & \{0,1\} \\
\max & & & & & \frac{5}{6}\, y_3 & & \\
\text{subject to} & 5\, y_1 & + & 4\, y_2 & + & 6\, y_3 & \leq & 4 \\
& y_1, & & y_2, & & y_3 & \in & \{0,1\}
\end{array}$$

  with $z(\text{ILP}_{\text{CSP}}) = n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{5}{6} - z(\text{ILP}^2_{\text{CSP}})$.

  b) **Greedy heuristic**
  The indices are scanned by non-increasing ratio q_j / w_j, starting from the residual right-hand side 4:

j	q_j	w_j	q_j/w_j	Decision	Residual
3	$\frac{5}{6}$	6	$\frac{5}{36}$	skip	4
1	0	5	0	$q_j = 0$	4
2	0	4	0	$q_j = 0$	4

The greedy solution is $\bar{\boldsymbol{y}} = (0, 0, 0)$, with value 0. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{1, 2, 3\}$, of total coefficient $15 > 10$ and cost $\frac{5}{6} - 0 = \frac{5}{6} < 1$. Hence the cover inequality

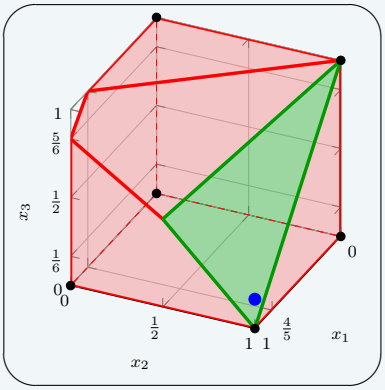
$$x_1 + x_2 + x_3 \leq 2$$

is violated by $\tilde{\boldsymbol{x}}$: $1 + 1 + \frac{1}{6} - 2 = \frac{1}{6}$.

c) **Optimality check**

No index with $q_j > 0$ satisfies the constraint with right-hand side 4 alone, hence every feasible solution of $\text{ILP}^2_{\text{CSP}}$ has value 0. Hence $z(\text{ILP}^2_{\text{CSP}}) = 0$, the greedy solution is optimal, $z(\text{ILP}_{\text{CSP}}) = \frac{5}{6} < 1$, and the maximum violation of a cover inequality is $\frac{1}{6}$.

The cover $\bar{C}$ is not minimal: removing an index with $q_j = 0$ gives the minimal cover $\{1, 3\}$, with the same cost $\frac{5}{6}$. Its cover inequality $x_1 + x_3 \leq 1$ has the same violation $\frac{1}{6}$ and is stronger.



Case (a): the LP relaxation and the cover inequality $x_1 + x_2 + x_3 \leq 2$.

- **Case (b): $\tilde{\boldsymbol{x}} = \left(\frac{4}{5}, 0, 1\right)$**

a) **Separation problem**

The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(\frac{1}{5}, 1, 0\right)$. The separation problem ILP_{CSP} (left) and, with $y_j = 1 - z_j$, the complementary problem $\text{ILP}^2_{\text{CSP}}$ (right) are:

$$\begin{aligned}
\min \quad & \tfrac{1}{5} z_1 + z_2 \\
\text{subject to} \quad & 5 z_1 + 4 z_2 + 6 z_3 \geq 11 \\
& z_1, \quad z_2, \quad z_3 \in \{0, 1\}
\end{aligned}$$

$$\begin{aligned}
\max \quad & \tfrac{1}{5} y_1 + y_2 \\
\text{subject to} \quad & 5 y_1 + 4 y_2 + 6 y_3 \leq 4 \\
& y_1, \quad y_2, \quad y_3 \in \{0, 1\}
\end{aligned}$$

with $z(\text{ILP}_{\text{CSP}}) = n - \sum_{j \in \mathscr{J}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{6}{5} - z(\text{ILP}^2_{\text{CSP}})$.

b) **Greedy heuristic**
The indices are scanned by non-increasing ratio q_j/w_j, starting from the residual right-hand side 4:

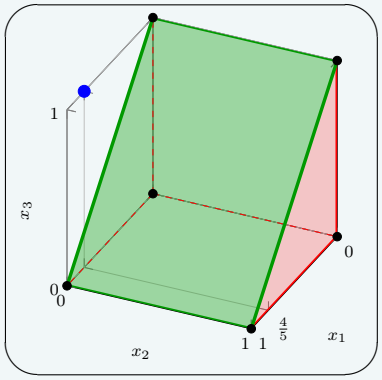
j	q_j	w_j	q_j/w_j	Decision	Residual
2	1	4	$\frac{1}{4}$	take	0
1	$\frac{1}{5}$	5	$\frac{1}{25}$	skip	0
3	0	6	0	$q_j = 0$	0

The greedy solution is $\bar{\boldsymbol{y}} = (0, 1, 0)$, with value 1. The corresponding cover is $\bar{C} = \{j \in \mathscr{J} : \bar{y}_j = 0\} = \{1, 3\}$, of total coefficient $11 > 10$ and cost $\frac{6}{5} - 1 = \frac{1}{5} < 1$. Hence the cover inequality

$$x_1 + x_3 \leq 1$$

is violated by $\tilde{\boldsymbol{x}}$: $\frac{4}{5} + 1 - 1 = \frac{4}{5}$.

c) **Optimality check**
Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{1, 2\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 4 are: $\{2\}$ (value 1). Hence $z(\text{ILP}^2_{\text{CSP}}) = 1$, the greedy solution is optimal, $z(\text{ILP}_{\text{CSP}}) = \frac{1}{5} < 1$, and the maximum violation of a cover inequality is $\frac{4}{5}$.



Case (b): the LP relaxation and the cover inequality $x_1 + x_3 \leq 1$.

- **Case (c):** $\tilde{\boldsymbol{x}} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$

a) **Separation problem**
The costs $q_j = 1 - \tilde{x}_j$ are $q = \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right)$. The separation problem ILP_{CSP} (left) and, with $y_j = 1 - z_j$, the complementary problem $\text{ILP}^2_{\text{CSP}}$ (right) are:

$$\begin{aligned} \min \quad & \tfrac{2}{3} z_1 + \tfrac{2}{3} z_2 + \tfrac{2}{3} z_3 \\ \text{subject to} \quad & 5 z_1 + 4 z_2 + 6 z_3 \geq 11 \\ & z_1, \quad z_2, \quad z_3 \in \{0, 1\} \end{aligned}$$

$$\begin{array}{rrcrcrcl} \max & \frac{2}{3}\,y_1 & + & \frac{2}{3}\,y_2 & + & \frac{2}{3}\,y_3 & & \\ \text{subject to} & 5\,y_1 & + & 4\,y_2 & + & 6\,y_3 & \leq & 4 \\ & y_1, & & y_2, & & y_3 & \in & \{0,1\} \end{array}$$

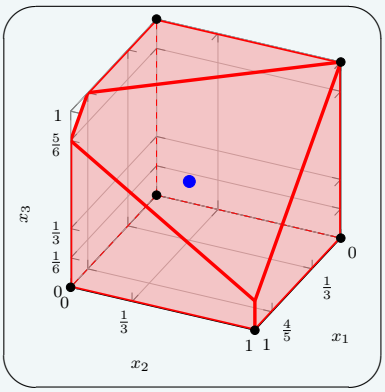
with $z(\mathrm{ILP}_{\mathrm{CSP}}) = n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}}) = 2 - z(\mathrm{ILP}^2_{\mathrm{CSP}})$.

b) **Greedy heuristic**
The indices are scanned by non-increasing ratio q_j/w_j, starting from the residual right-hand side 4:

j	q_j	w_j	q_j/w_j	Decision	Residual
2	$\frac{2}{3}$	4	$\frac{1}{6}$	take	0
1	$\frac{2}{3}$	5	$\frac{2}{15}$	skip	0
3	$\frac{2}{3}$	6	$\frac{1}{9}$	skip	0

The greedy solution is $\bar{\boldsymbol{y}} = (0,1,0)$, with value $\frac{2}{3}$. The corresponding cover is $\bar{C} = \{j \in \mathscr{I} : \bar{y}_j = 0\} = \{1,3\}$, with cost $2 - \frac{2}{3} = \frac{4}{3} \geq 1$: its cover inequality is not violated by $\tilde{\boldsymbol{x}}$.

c) **Optimality check**
Indices with $q_j = 0$ do not change the value of a solution. The non-empty subsets of the indices $\{1,2,3\}$ with $q_j > 0$ that satisfy the constraint with right-hand side 4 are: $\{2\}$ (value $\frac{2}{3}$). Hence $z(\mathrm{ILP}^2_{\mathrm{CSP}}) = \frac{2}{3}$, the greedy solution is optimal, and $z(\mathrm{ILP}_{\mathrm{CSP}}) = \frac{4}{3} \geq 1$: **no** cover inequality is violated by $\tilde{\boldsymbol{x}}$.



Case (c): the LP relaxation; no cover inequality is violated by the point.

5.3 Closure and separation

Exercise 5.11 — Cover closure by separation ($P1$)

Consider a binary problem with one constraint $\sum_j w_j x_j \leq c$, with positive integer coefficients w_j and right-hand side $c = 9$, on the binary variables indexed by the set $\mathscr{I} = \{1, 2, 3, 4\}$ of $n = 4$ variables with the following objective coefficients and constraint coefficients:

j	1	2	3	4
p_j	22	33	9	13
w_j	5	9	3	6

The associated LP is:

$$\begin{array}{rrcrcrcrcl} \max & 22\,x_1 & + & 33\,x_2 & + & 9\,x_3 & + & 13\,x_4 & & \\ \text{subject to} & 5\,x_1 & + & 9\,x_2 & + & 3\,x_3 & + & 6\,x_4 & \leq & 9 \\ & x_1 & & & & & & & \leq & 1 \\ & & & x_2 & & & & & \leq & 1 \\ & & & & & x_3 & & & \leq & 1 \\ & & & & & & & x_4 & \leq & 1 \\ & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0 \end{array}$$

An optimal solution $\tilde{\boldsymbol{x}}$ of LP and its value are:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = \frac{4}{9}, \quad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 0 \qquad \text{and} \qquad z(\text{LP}) = \frac{110}{3}.$$

Questions

1. Starting from $\tilde{\boldsymbol{x}}$, separate the cover inequalities by solving the complementary problem $\text{ILP}^2_{\text{CSP}}$ to optimality. Add the maximally violated cover inequality to the LP, solve it again, and repeat until no cover inequality is violated. If $\text{ILP}^2_{\text{CSP}}$ has several optimal solutions, choose the one that selects the largest number of variables.
2. What is the optimal value over the cover closure? Is the final solution integer?

The **cover closure** is the polyhedron obtained by adding to LP all the cover inequalities $\sum_{j \in C} x_j \leq |C| - 1$, $C \in \mathcal{C}$. Given a point $\tilde{\boldsymbol{x}}$, let $\tilde{y}_1, \ldots, \tilde{y}_n$ be an optimal solution of the complementary problem

$$(\text{ILP}^2_{\text{CSP}}) \qquad \max \Big\{ \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j)\, y_j : \sum_{j \in \mathscr{I}} w_j\, y_j \leq \sum_{j \in \mathscr{I}} w_j - (c+1),\ y_j \in \{0,1\}\ \forall j \in \mathscr{I} \Big\}.$$

If $n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) < 1$, the cover $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\}$ is associated with the cover inequality maximally violated by $\tilde{\boldsymbol{x}}$; otherwise, no cover inequality is violated by $\tilde{\boldsymbol{x}}$.

Solution

Separation of the cover inequalities with the complementary problem; in all the iterations, the right-hand side of $\text{ILP}^2_{\text{CSP}}$ is $\sum_{j\in\mathscr{I}} w_j - (c+1) = 23 - 10 = 13$.

- **Iteration 1**
  The optimal solution of LP is $\tilde{\boldsymbol{x}} = \left(1, \frac{4}{9}, 0, 0\right)$, with value $z(\text{LP}) = \frac{110}{3}$.
  The costs are $q = \left(0, \frac{5}{9}, 1, 1\right)$, the constant term is $n - \sum_{j\in\mathscr{I}} \tilde{x}_j = \frac{23}{9}$, and the complementary problem is:

$$\begin{array}{rccccccccc} \max & & & \frac{5}{9}\,y_2 & + & y_3 & + & y_4 & & \\ \text{subject to} & 5\,y_1 & + & 9\,y_2 & + & 3\,y_3 & + & 6\,y_4 & \leq & 13 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

  The maximal subsets of indices that satisfy the constraint with right-hand side 13 are: $\{1,3\}$ (value 1), $\{1,4\}$ (value 1), $\{2,3\}$ (value $\frac{14}{9}$), $\{3,4\}$ (value 2). An optimal solution of $\text{ILP}^2_{\text{CSP}}$ is $\tilde{y}_1 = 0,\ \tilde{y}_2 = 0,\ \tilde{y}_3 = 1,\ \tilde{y}_4 = 1$, with value $z(\text{ILP}^2_{\text{CSP}}) = 2$:

$$n - \sum_{j\in\mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{23}{9} - 2 = \frac{5}{9} < 1.$$

  Accordingly, $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\} = \{1,2\}$ and the cover inequality

$$x_1 + x_2 \leq 1$$

  is maximally violated by $\tilde{\boldsymbol{x}}$: $1 + \frac{4}{9} - 1 = \frac{4}{9}$.

- **Iteration 2**
  Adding the cover inequality $x_1 + x_2 \leq 1$, we obtain the following LP, denoted LP_1:

$$\begin{array}{rccccccccc} \max & 22\,x_1 & + & 33\,x_2 & + & 9\,x_3 & + & 13\,x_4 & & \\ \text{subject to} & 5\,x_1 & + & 9\,x_2 & + & 3\,x_3 & + & 6\,x_4 & \leq & 9 \\ & x_1 & + & x_2 & & & & & \leq & 1 \\ & x_1 & & & & & & & \leq & 1 \\ & & & x_2 & & & & & \leq & 1 \\ & & & & & x_3 & & & \leq & 1 \\ & & & & & & & x_4 & \leq & 1 \\ & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0 \end{array}$$

  An optimal solution of LP_1 is $\tilde{\boldsymbol{x}} = \left(\frac{3}{4}, \frac{1}{4}, 1, 0\right)$, with value $z(\text{LP}_1) = \frac{135}{4}$.
  The costs are $q = \left(\frac{1}{4}, \frac{3}{4}, 0, 1\right)$, the constant term is $n - \sum_{j\in\mathscr{I}} \tilde{x}_j = 2$, and the complementary problem is:

$$\begin{array}{rccccccccc} \max & \frac{1}{4}\,y_1 & + & \frac{3}{4}\,y_2 & & & + & y_4 & & \\ \text{subject to} & 5\,y_1 & + & 9\,y_2 & + & 3\,y_3 & + & 6\,y_4 & \leq & 13 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

The maximal subsets of indices that satisfy the constraint with right-hand side 13 are: $\{1,3\}$ (value $\frac{1}{4}$), $\{1,4\}$ (value $\frac{5}{4}$), $\{2,3\}$ (value $\frac{3}{4}$), $\{3,4\}$ (value 1). An optimal solution of $\text{ILP}^2_{\text{CSP}}$ is $\tilde{y}_1 = 1,\ \tilde{y}_2 = 0,\ \tilde{y}_3 = 0,\ \tilde{y}_4 = 1$, with value $z(\text{ILP}^2_{\text{CSP}}) = \frac{5}{4}$:

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = 2 - \frac{5}{4} = \frac{3}{4} < 1.$$

Accordingly, $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\} = \{2,3\}$ and the cover inequality

$$x_2 + x_3 \leq 1$$

is maximally violated by $\tilde{\boldsymbol{x}}$: $\frac{1}{4} + 1 - 1 = \frac{1}{4}$.

- **Iteration 3**

  Adding the cover inequality $x_2 + x_3 \leq 1$, we obtain the following LP, denoted LP_2:

$$\begin{array}{rrcrcrcrcl}
\max & 22\,x_1 & + & 33\,x_2 & + & 9\,x_3 & + & 13\,x_4 & & \\
\text{subject to} & 5\,x_1 & + & 9\,x_2 & + & 3\,x_3 & + & 6\,x_4 & \leq & 9 \\
& x_1 & + & x_2 & & & & & \leq & 1 \\
& & & x_2 & + & x_3 & & & \leq & 1 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}$$

  An optimal solution of LP_2 is $\tilde{\boldsymbol{x}} = \left(1, 0, 1, \frac{1}{6}\right)$, with value $z(\text{LP}_2) = \frac{199}{6}$.

  The costs are $q = \left(0, 1, 0, \frac{5}{6}\right)$, the constant term is $n - \sum_{j \in \mathscr{I}} \tilde{x}_j = \frac{11}{6}$, and the complementary problem is:

$$\begin{array}{rrcrcrcrcl}
\max & & & y_2 & & & + & \frac{5}{6}\,y_4 & & \\
\text{subject to} & 5\,y_1 & + & 9\,y_2 & + & 3\,y_3 & + & 6\,y_4 & \leq & 13 \\
& y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\}
\end{array}$$

  The maximal subsets of indices that satisfy the constraint with right-hand side 13 are: $\{1,3\}$ (value 0), $\{1,4\}$ (value $\frac{5}{6}$), $\{2,3\}$ (value 1), $\{3,4\}$ (value $\frac{5}{6}$). An optimal solution of $\text{ILP}^2_{\text{CSP}}$ is $\tilde{y}_1 = 0,\ \tilde{y}_2 = 1,\ \tilde{y}_3 = 1,\ \tilde{y}_4 = 0$, with value $z(\text{ILP}^2_{\text{CSP}}) = 1$:

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{11}{6} - 1 = \frac{5}{6} < 1.$$

  Accordingly, $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\} = \{1,4\}$ and the cover inequality

$$x_1 + x_4 \leq 1$$

  is maximally violated by $\tilde{\boldsymbol{x}}$: $1 + \frac{1}{6} - 1 = \frac{1}{6}$.

- **Iteration 4**
  Adding the cover inequality $x_1 + x_4 \leq 1$, we obtain the following LP, denoted LP_3:

$$\begin{array}{rrcrcrcrcl}
\max & 22\,x_1 & + & 33\,x_2 & + & 9\,x_3 & + & 13\,x_4 & & \\
\text{subject to} & 5\,x_1 & + & 9\,x_2 & + & 3\,x_3 & + & 6\,x_4 & \leq & 9 \\
& x_1 & + & x_2 & & & & & \leq & 1 \\
& & & x_2 & + & x_3 & & & \leq & 1 \\
& x_1 & & & & & + & x_4 & \leq & 1 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}$$

  An optimal solution of LP_3 is $\tilde{\boldsymbol{x}} = \left(\frac{6}{7}, \frac{1}{7}, \frac{6}{7}, \frac{1}{7}\right)$, with value $z(\text{LP}_3) = \frac{232}{7}$.
  The costs are $q = \left(\frac{1}{7}, \frac{6}{7}, \frac{1}{7}, \frac{6}{7}\right)$, the constant term is $n - \sum_{j \in \mathscr{I}} \tilde{x}_j = 2$, and the complementary problem is:

$$\begin{array}{rrcrcrcrcl}
\max & \frac{1}{7}\,y_1 & + & \frac{6}{7}\,y_2 & + & \frac{1}{7}\,y_3 & + & \frac{6}{7}\,y_4 & & \\
\text{subject to} & 5\,y_1 & + & 9\,y_2 & + & 3\,y_3 & + & 6\,y_4 & \leq & 13 \\
& y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\}
\end{array}$$

  The maximal subsets of indices that satisfy the constraint with right-hand side 13 are: $\{1,3\}$ (value $\frac{2}{7}$), $\{1,4\}$ (value 1), $\{2,3\}$ (value 1), $\{3,4\}$ (value 1). The optimal value of $\text{ILP}^2_{\text{CSP}}$ is $z(\text{ILP}^2_{\text{CSP}}) = 1$:

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = 2 - 1 = 1 \geq 1.$$

  No cover inequality is violated by $\tilde{\boldsymbol{x}} = \left(\frac{6}{7}, \frac{1}{7}, \frac{6}{7}, \frac{1}{7}\right)$: the separation stops after 3 cover inequalities. The optimal value over the cover closure is

$$z(\text{LP}^{\text{cov}}) = z(\text{LP}_3) = \frac{232}{7} < z(\text{LP}) = \frac{110}{3},$$

  and $\tilde{\boldsymbol{x}}$ is **fractional**: the cover closure does not coincide with the convex hull of the integer solutions.

Exercise 5.12 — Cover closure by separation ($P2$)

Consider a binary problem with one constraint $\sum_j w_j x_j \leq c$, with positive integer coefficients w_j and right-hand side $c = 17$, on the binary variables indexed by the set $\mathscr{I} = \{1,2,3,4\}$ of $n = 4$ variables with the following objective coefficients and constraint

coefficients:

j	1	2	3	4
p_j	29	29	27	21
w_j	5	10	10	8

The associated LP is:

$$\begin{array}{rrcrcrcrcl} \max & 29\,x_1 & + & 29\,x_2 & + & 27\,x_3 & + & 21\,x_4 & & \\ \text{subject to} & 5\,x_1 & + & 10\,x_2 & + & 10\,x_3 & + & 8\,x_4 & \leq & 17 \\ & x_1 & & & & & & & \leq & 1 \\ & & & x_2 & & & & & \leq & 1 \\ & & & & & x_3 & & & \leq & 1 \\ & & & & & & & x_4 & \leq & 1 \\ & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0 \end{array}$$

An optimal solution $\tilde{\boldsymbol{x}}$ of LP and its value are:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_3 = \frac{1}{5}, \quad \tilde{x}_4 = 0 \qquad \text{and} \qquad z(\text{LP}) = \frac{317}{5}.$$

Questions

1. Starting from $\tilde{\boldsymbol{x}}$, separate the cover inequalities by solving the complementary problem $\text{ILP}^2_{\text{CSP}}$ to optimality. Add the maximally violated cover inequality to the LP, solve it again, and repeat until no cover inequality is violated. If $\text{ILP}^2_{\text{CSP}}$ has several optimal solutions, choose the one that selects the largest number of variables.
2. What is the optimal value over the cover closure? Is the final solution integer?

The **cover closure** is the polyhedron obtained by adding to LP all the cover inequalities $\sum_{j\in C} x_j \leq |C|-1$, $C \in \mathcal{C}$. Given a point $\tilde{\boldsymbol{x}}$, let $\tilde{y}_1, \ldots, \tilde{y}_n$ be an optimal solution of the complementary problem

$$(\text{ILP}^2_{\text{CSP}}) \qquad \max \left\{ \sum_{j\in\mathscr{I}} (1-\tilde{x}_j)\, y_j : \sum_{j\in\mathscr{I}} w_j\, y_j \leq \sum_{j\in\mathscr{I}} w_j - (c+1),\ y_j \in \{0,1\}\ \forall j \in \mathscr{I} \right\}.$$

If $n - \sum_{j\in\mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) < 1$, the cover $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\}$ is associated with the cover inequality maximally violated by $\tilde{\boldsymbol{x}}$; otherwise, no cover inequality is violated by $\tilde{\boldsymbol{x}}$.

Solution

Separation of the cover inequalities with the complementary problem; in all the iterations, the right-hand side of $\text{ILP}^2_{\text{CSP}}$ is $\sum_{j\in\mathscr{I}} w_j - (c+1) = 33 - 18 = 15$.

- **Iteration 1**

The optimal solution of LP is $\tilde{\boldsymbol{x}} = \left(1, 1, \frac{1}{5}, 0\right)$, with value $z(\text{LP}) = \frac{317}{5}$.
The costs are $q = \left(0, 0, \frac{4}{5}, 1\right)$, the constant term is $n - \sum_{j \in \mathscr{I}} \tilde{x}_j = \frac{9}{5}$, and the complementary problem is:

$$\begin{array}{rccccccccc} \max & & & & & \frac{4}{5}\, y_3 & + & y_4 & & \\ \text{subject to} & 5\, y_1 & + & 10\, y_2 & + & 10\, y_3 & + & 8\, y_4 & \leq & 15 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

The maximal subsets of indices that satisfy the constraint with right-hand side 15 are: $\{1,2\}$ (value 0), $\{1,3\}$ (value $\frac{4}{5}$), $\{1,4\}$ (value 1). An optimal solution of $\text{ILP}^2_{\text{CSP}}$ is $\tilde{y}_1 = 1,\ \tilde{y}_2 = 0,\ \tilde{y}_3 = 0,\ \tilde{y}_4 = 1$, with value $z(\text{ILP}^2_{\text{CSP}}) = 1$:

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{9}{5} - 1 = \frac{4}{5} < 1.$$

Accordingly, $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\} = \{2, 3\}$ and the cover inequality

$$x_2 + x_3 \leq 1$$

is maximally violated by $\tilde{\boldsymbol{x}}$: $1 + \frac{1}{5} - 1 = \frac{1}{5}$.

- **Iteration 2**
  Adding the cover inequality $x_2 + x_3 \leq 1$, we obtain the following LP, denoted LP_1:

$$\begin{array}{rccccccccc} \max & 29\, x_1 & + & 29\, x_2 & + & 27\, x_3 & + & 21\, x_4 & & \\ \text{subject to} & 5\, x_1 & + & 10\, x_2 & + & 10\, x_3 & + & 8\, x_4 & \leq & 17 \\ & & & x_2 & + & x_3 & & & \leq & 1 \\ & x_1 & & & & & & & \leq & 1 \\ & & & x_2 & & & & & \leq & 1 \\ & & & & & x_3 & & & \leq & 1 \\ & & & & & & & x_4 & \leq & 1 \\ & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0 \end{array}$$

An optimal solution of LP_1 is $\tilde{\boldsymbol{x}} = \left(1, 1, 0, \frac{1}{4}\right)$, with value $z(\text{LP}_1) = \frac{253}{4}$.
The costs are $q = \left(0, 0, 1, \frac{3}{4}\right)$, the constant term is $n - \sum_{j \in \mathscr{I}} \tilde{x}_j = \frac{7}{4}$, and the complementary problem is:

$$\begin{array}{rccccccccc} \max & & & & & y_3 & + & \frac{3}{4}\, y_4 & & \\ \text{subject to} & 5\, y_1 & + & 10\, y_2 & + & 10\, y_3 & + & 8\, y_4 & \leq & 15 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

The maximal subsets of indices that satisfy the constraint with right-hand side 15 are: $\{1,2\}$ (value 0), $\{1,3\}$ (value 1), $\{1,4\}$ (value $\frac{3}{4}$). An optimal solution of $\text{ILP}^2_{\text{CSP}}$ is $\tilde{y}_1 = 1,\ \tilde{y}_2 = 0,\ \tilde{y}_3 = 1,\ \tilde{y}_4 = 0$, with value $z(\text{ILP}^2_{\text{CSP}}) = 1$:

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{7}{4} - 1 = \frac{3}{4} < 1.$$

Accordingly, $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\} = \{2, 4\}$ and the cover inequality

$$x_2 + x_4 \leq 1$$

is maximally violated by $\tilde{\boldsymbol{x}}$: $1 + \frac{1}{4} - 1 = \frac{1}{4}$.

- **Iteration 3**
  Adding the cover inequality $x_2 + x_4 \leq 1$, we obtain the following LP, denoted LP_2:

$$\begin{array}{rrcrcrcrcl}
\max & 29\,x_1 & + & 29\,x_2 & + & 27\,x_3 & + & 21\,x_4 & & \\
\text{subject to} & 5\,x_1 & + & 10\,x_2 & + & 10\,x_3 & + & 8\,x_4 & \leq & 17 \\
 & & & x_2 & + & x_3 & & & \leq & 1 \\
 & & & x_2 & & & + & x_4 & \leq & 1 \\
 & x_1 & & & & & & & \leq & 1 \\
 & & & x_2 & & & & & \leq & 1 \\
 & & & & & x_3 & & & \leq & 1 \\
 & & & & & & & x_4 & \leq & 1 \\
 & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}$$

  An optimal solution of LP_2 is $\tilde{\boldsymbol{x}} = \left(1, \frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right)$, with value $z(\text{LP}_2) = \frac{251}{4}$.
  The costs are $q = \left(0, \frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right)$, the constant term is $n - \sum_{j \in \mathscr{I}} \tilde{x}_j = \frac{7}{4}$, and the complementary problem is:

$$\begin{array}{rrcrcrcrcl}
\max & & & \frac{1}{4}\,y_2 & + & \frac{3}{4}\,y_3 & + & \frac{3}{4}\,y_4 & & \\
\text{subject to} & 5\,y_1 & + & 10\,y_2 & + & 10\,y_3 & + & 8\,y_4 & \leq & 15 \\
 & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0, 1\}
\end{array}$$

  The maximal subsets of indices that satisfy the constraint with right-hand side 15 are: $\{1, 2\}$ (value $\frac{1}{4}$), $\{1, 3\}$ (value $\frac{3}{4}$), $\{1, 4\}$ (value $\frac{3}{4}$). The optimal value of $\text{ILP}^2_{\text{CSP}}$ is $z(\text{ILP}^2_{\text{CSP}}) = \frac{3}{4}$:

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{7}{4} - \frac{3}{4} = 1 \geq 1.$$

  No cover inequality is violated by $\tilde{\boldsymbol{x}} = \left(1, \frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right)$: the separation stops after 2 cover inequalities. The optimal value over the cover closure is

$$z(\text{LP}^{\text{cov}}) = z(\text{LP}_2) = \frac{251}{4} < z(\text{LP}) = \frac{317}{5},$$

  and $\tilde{\boldsymbol{x}}$ is **fractional**: the cover closure does not coincide with the convex hull of the integer solutions.

Exercise 5.13 — Cover closure by separation ($P3$)

Consider a binary problem with one constraint $\sum_j w_j x_j \leq c$, with positive integer coefficients w_j and right-hand side $c = 15$, on the binary variables indexed by the set $\mathscr{I} = \{1, 2, 3, 4\}$ of $n = 4$ variables with the following objective coefficients and constraint coefficients:

j	1	2	3	4
p_j	30	8	9	6
w_j	5	2	9	9

The associated LP is:

$$
\begin{array}{rrcrcrcrcl}
\max & 30\,x_1 & + & 8\,x_2 & + & 9\,x_3 & + & 6\,x_4 & & \\
\text{subject to} & 5\,x_1 & + & 2\,x_2 & + & 9\,x_3 & + & 9\,x_4 & \leq & 15 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}
$$

An optimal solution $\tilde{\boldsymbol{x}}$ of LP and its value are:

$$
\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_3 = \frac{8}{9}, \quad \tilde{x}_4 = 0 \qquad \text{and} \qquad z(\text{LP}) = 46.
$$

Questions

1. Starting from $\tilde{\boldsymbol{x}}$, separate the cover inequalities by solving the complementary problem $\text{ILP}^2_{\text{CSP}}$ to optimality. Add the maximally violated cover inequality to the LP, solve it again, and repeat until no cover inequality is violated. If $\text{ILP}^2_{\text{CSP}}$ has several optimal solutions, choose the one that selects the largest number of variables.
2. What is the optimal value over the cover closure? Is the final solution integer?

The **cover closure** is the polyhedron obtained by adding to LP all the cover inequalities $\sum_{j \in C} x_j \leq |C| - 1$, $C \in \mathcal{C}$. Given a point $\tilde{\boldsymbol{x}}$, let $\tilde{y}_1, \ldots, \tilde{y}_n$ be an optimal solution of the complementary problem

$$
(\text{ILP}^2_{\text{CSP}}) \qquad \max \Big\{ \sum_{j \in \mathscr{I}} (1 - \tilde{x}_j)\, y_j : \sum_{j \in \mathscr{I}} w_j\, y_j \leq \sum_{j \in \mathscr{I}} w_j - (c+1),\ y_j \in \{0,1\}\ \forall j \in \mathscr{I} \Big\}.
$$

If $n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) < 1$, the cover $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\}$ is associated with the cover inequality maximally violated by $\tilde{\boldsymbol{x}}$; otherwise, no cover inequality is violated by $\tilde{\boldsymbol{x}}$.

Solution

Separation of the cover inequalities with the complementary problem; in all the iterations, the right-hand side of $\text{ILP}^2_{\text{CSP}}$ is $\sum_{j\in\mathscr{I}} w_j - (c+1) = 25 - 16 = 9$.

- **Iteration 1**
  The optimal solution of LP is $\tilde{\boldsymbol{x}} = \left(1, 1, \frac{8}{9}, 0\right)$, with value $z(\text{LP}) = 46$.
  The costs are $q = \left(0, 0, \frac{1}{9}, 1\right)$, the constant term is $n - \sum_{j\in\mathscr{I}} \tilde{x}_j = \frac{10}{9}$, and the complementary problem is:

$$\begin{array}{rccccccccc} \max & & & & & \frac{1}{9}\,y_3 & + & y_4 & & \\ \text{subject to} & 5\,y_1 & + & 2\,y_2 & + & 9\,y_3 & + & 9\,y_4 & \leq & 9 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

  The maximal subsets of indices that satisfy the constraint with right-hand side 9 are: $\{3\}$ (value $\frac{1}{9}$), $\{4\}$ (value 1), $\{1,2\}$ (value 0). An optimal solution of $\text{ILP}^2_{\text{CSP}}$ is $\tilde{y}_1 = 0,\ \tilde{y}_2 = 0,\ \tilde{y}_3 = 0,\ \tilde{y}_4 = 1$, with value $z(\text{ILP}^2_{\text{CSP}}) = 1$:

$$n - \sum_{j\in\mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{10}{9} - 1 = \frac{1}{9} < 1.$$

  Accordingly, $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\} = \{1, 2, 3\}$ and the cover inequality

$$x_1 + x_2 + x_3 \leq 2$$

  is maximally violated by $\tilde{\boldsymbol{x}}$: $1 + 1 + \frac{8}{9} - 2 = \frac{8}{9}$.

- **Iteration 2**
  Adding the cover inequality $x_1 + x_2 + x_3 \leq 2$, we obtain the following LP, denoted LP_1:

$$\begin{array}{rccccccccc} \max & 30\,x_1 & + & 8\,x_2 & + & 9\,x_3 & + & 6\,x_4 & & \\ \text{subject to} & 5\,x_1 & + & 2\,x_2 & + & 9\,x_3 & + & 9\,x_4 & \leq & 15 \\ & x_1 & + & x_2 & + & x_3 & & & \leq & 2 \\ & x_1 & & & & & & & \leq & 1 \\ & & & x_2 & & & & & \leq & 1 \\ & & & & & x_3 & & & \leq & 1 \\ & & & & & & & x_4 & \leq & 1 \\ & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0 \end{array}$$

  An optimal solution of LP_1 is $\tilde{\boldsymbol{x}} = \left(1, 1, 0, \frac{8}{9}\right)$, with value $z(\text{LP}_1) = \frac{130}{3}$.
  The costs are $q = \left(0, 0, 1, \frac{1}{9}\right)$, the constant term is $n - \sum_{j\in\mathscr{I}} \tilde{x}_j = \frac{10}{9}$, and the complementary problem is:

$$\begin{array}{rccccccccc} \max & & & & & y_3 & + & \frac{1}{9}\,y_4 & & \\ \text{subject to} & 5\,y_1 & + & 2\,y_2 & + & 9\,y_3 & + & 9\,y_4 & \leq & 9 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

The maximal subsets of indices that satisfy the constraint with right-hand side 9 are: $\{3\}$ (value 1), $\{4\}$ (value $\frac{1}{9}$), $\{1,2\}$ (value 0). An optimal solution of $\text{ILP}^2_{\text{CSP}}$ is $\tilde{y}_1 = 0,\ \tilde{y}_2 = 0,\ \tilde{y}_3 = 1,\ \tilde{y}_4 = 0$, with value $z(\text{ILP}^2_{\text{CSP}}) = 1$:

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{10}{9} - 1 = \frac{1}{9} < 1.$$

Accordingly, $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\} = \{1,2,4\}$ and the cover inequality

$$x_1 + x_2 + x_4 \leq 2$$

is maximally violated by $\tilde{\boldsymbol{x}}$: $1 + 1 + \frac{8}{9} - 2 = \frac{8}{9}$.

- **Iteration 3**
  Adding the cover inequality $x_1 + x_2 + x_4 \leq 2$, we obtain the following LP, denoted LP_2:

$$\begin{array}{rrcrcrcrcl}
\max & 30\,x_1 & + & 8\,x_2 & + & 9\,x_3 & + & 6\,x_4 & & \\
\text{subject to} & 5\,x_1 & + & 2\,x_2 & + & 9\,x_3 & + & 9\,x_4 & \leq & 15 \\
& x_1 & + & x_2 & + & x_3 & & & \leq & 2 \\
& x_1 & + & x_2 & & & + & x_4 & \leq & 2 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}$$

An optimal solution of LP_2 is $\tilde{\boldsymbol{x}} = \left(1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$, with value $z(\text{LP}_2) = \frac{83}{2}$.

The costs are $q = \left(0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$, the constant term is $n - \sum_{j \in \mathscr{I}} \tilde{x}_j = \frac{3}{2}$, and the complementary problem is:

$$\begin{array}{rrcrcrcrcl}
\max & & & \frac{1}{2}\,y_2 & + & \frac{1}{2}\,y_3 & + & \frac{1}{2}\,y_4 & & \\
\text{subject to} & 5\,y_1 & + & 2\,y_2 & + & 9\,y_3 & + & 9\,y_4 & \leq & 9 \\
& y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\}
\end{array}$$

The maximal subsets of indices that satisfy the constraint with right-hand side 9 are: $\{3\}$ (value $\frac{1}{2}$), $\{4\}$ (value $\frac{1}{2}$), $\{1,2\}$ (value $\frac{1}{2}$). The optimal value of $\text{ILP}^2_{\text{CSP}}$ is $z(\text{ILP}^2_{\text{CSP}}) = \frac{1}{2}$:

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = \frac{3}{2} - \frac{1}{2} = 1 \geq 1.$$

No cover inequality is violated by $\tilde{\boldsymbol{x}} = \left(1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$: the separation stops after 2 cover inequalities. The optimal value over the cover closure is

$$z(\mathrm{LP}^{\mathrm{cov}}) = z(\mathrm{LP}_2) = \frac{83}{2} < z(\mathrm{LP}) = 46,$$

and $\tilde{\boldsymbol{x}}$ is **fractional**: the cover closure does not coincide with the convex hull of the integer solutions.

Exercise 5.14 — Cover closure by separation ($P4$)

Consider a binary problem with one constraint $\sum_j w_j x_j \leq c$, with positive integer coefficients w_j and right-hand side $c = 9$, on the binary variables indexed by the set $\mathscr{I} = \{1, 2, 3, 4\}$ of $n = 4$ variables with the following objective coefficients and constraint coefficients:

j	1	2	3	4
p_j	31	13	5	9
w_j	8	6	3	8

The associated LP is:

$$\begin{array}{rrcrcrcrcl}
\max & 31\,x_1 & + & 13\,x_2 & + & 5\,x_3 & + & 9\,x_4 & & \\
\text{subject to} & 8\,x_1 & + & 6\,x_2 & + & 3\,x_3 & + & 8\,x_4 & \leq & 9 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}$$

An optimal solution $\tilde{\boldsymbol{x}}$ of LP and its value are:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = \frac{1}{6}, \quad \tilde{x}_3 = 0, \quad \tilde{x}_4 = 0 \qquad \text{and} \qquad z(\mathrm{LP}) = \frac{199}{6}.$$

Questions

1. Starting from $\tilde{\boldsymbol{x}}$, separate the cover inequalities by solving the complementary problem $\mathrm{ILP}^2_{\mathrm{CSP}}$ to optimality. Add the maximally violated cover inequality to the LP, solve it again, and repeat until no cover inequality is violated. If $\mathrm{ILP}^2_{\mathrm{CSP}}$ has several optimal solutions, choose the one that selects the largest number of variables.
2. What is the optimal value over the cover closure? Is the final solution integer?

The **cover closure** is the polyhedron obtained by adding to LP all the cover inequalities $\sum_{j\in C} x_j \le |C|-1$, $C \in \mathcal{C}$. Given a point $\tilde{\boldsymbol{x}}$, let $\tilde{y}_1, \dots, \tilde{y}_n$ be an optimal solution of the complementary problem

$$(\mathrm{ILP}^2_{\mathrm{CSP}}) \qquad \max\Big\{ \sum_{j\in\mathscr{I}} (1-\tilde{x}_j)\, y_j : \sum_{j\in\mathscr{I}} w_j\, y_j \le \sum_{j\in\mathscr{I}} w_j - (c+1),\ y_j \in \{0,1\}\ \forall j \in \mathscr{I} \Big\}.$$

If $n - \sum_{j\in\mathscr{I}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}}) < 1$, the cover $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\}$ is associated with the cover inequality maximally violated by $\tilde{\boldsymbol{x}}$; otherwise, no cover inequality is violated by $\tilde{\boldsymbol{x}}$.

Solution

Separation of the cover inequalities with the complementary problem; in all the iterations, the right-hand side of $\mathrm{ILP}^2_{\mathrm{CSP}}$ is $\sum_{j\in\mathscr{I}} w_j - (c+1) = 25 - 10 = 15$.

- **Iteration 1**
  The optimal solution of LP is $\tilde{\boldsymbol{x}} = \left(1, \frac{1}{6}, 0, 0\right)$, with value $z(\mathrm{LP}) = \frac{199}{6}$.
  The costs are $q = \left(0, \frac{5}{6}, 1, 1\right)$, the constant term is $n - \sum_{j\in\mathscr{I}} \tilde{x}_j = \frac{17}{6}$, and the complementary problem is:

$$\begin{array}{rrcrcrcrcl} \max & & & \frac{5}{6}\, y_2 & + & y_3 & + & y_4 & & \\ \text{subject to} & 8\, y_1 & + & 6\, y_2 & + & 3\, y_3 & + & 8\, y_4 & \le & 15 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

  The maximal subsets of indices that satisfy the constraint with right-hand side 15 are: $\{1,2\}$ (value $\frac{5}{6}$), $\{1,3\}$ (value 1), $\{2,3\}$ (value $\frac{11}{6}$), $\{2,4\}$ (value $\frac{11}{6}$), $\{3,4\}$ (value 2). An optimal solution of $\mathrm{ILP}^2_{\mathrm{CSP}}$ is $\tilde{y}_1 = 0$, $\tilde{y}_2 = 0$, $\tilde{y}_3 = 1$, $\tilde{y}_4 = 1$, with value $z(\mathrm{ILP}^2_{\mathrm{CSP}}) = 2$:

$$n - \sum_{j\in\mathscr{I}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}}) = \frac{17}{6} - 2 = \frac{5}{6} < 1.$$

  Accordingly, $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\} = \{1,2\}$ and the cover inequality

$$x_1 + x_2 \le 1$$

  is maximally violated by $\tilde{\boldsymbol{x}}$: $1 + \frac{1}{6} - 1 = \frac{1}{6}$.

- **Iteration 2**
  Adding the cover inequality $x_1 + x_2 \le 1$, we obtain the following LP, denoted LP_1:

$$\begin{array}{rrcrcrcrcl}
\max & 31\,x_1 & + & 13\,x_2 & + & 5\,x_3 & + & 9\,x_4 & & \\
\text{subject to} & 8\,x_1 & + & 6\,x_2 & + & 3\,x_3 & + & 8\,x_4 & \leq & 9 \\
& x_1 & + & x_2 & & & & & \leq & 1 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}$$

An optimal solution of LP_1 is $\tilde{\boldsymbol{x}} = \left(1, 0, \frac{1}{3}, 0\right)$, with value $z(\mathrm{LP}_1) = \frac{98}{3}$.

The costs are $q = \left(0, 1, \frac{2}{3}, 1\right)$, the constant term is $n - \sum_{j \in \mathscr{I}} \tilde{x}_j = \frac{8}{3}$, and the complementary problem is:

$$\begin{array}{rrcrcrcrcl}
\max & & & y_2 & + & \frac{2}{3}\,y_3 & + & y_4 & & \\
\text{subject to} & 8\,y_1 & + & 6\,y_2 & + & 3\,y_3 & + & 8\,y_4 & \leq & 15 \\
& y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\}
\end{array}$$

The maximal subsets of indices that satisfy the constraint with right-hand side 15 are: $\{1,2\}$ (value 1), $\{1,3\}$ (value $\frac{2}{3}$), $\{2,3\}$ (value $\frac{5}{3}$), $\{2,4\}$ (value 2), $\{3,4\}$ (value $\frac{5}{3}$). An optimal solution of $\mathrm{ILP}^2_{\mathrm{CSP}}$ is $\tilde{y}_1 = 0$, $\tilde{y}_2 = 1$, $\tilde{y}_3 = 0$, $\tilde{y}_4 = 1$, with value $z(\mathrm{ILP}^2_{\mathrm{CSP}}) = 2$:

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}}) = \frac{8}{3} - 2 = \frac{2}{3} < 1.$$

Accordingly, $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\} = \{1, 3\}$ and the cover inequality

$$x_1 + x_3 \leq 1$$

is maximally violated by $\tilde{\boldsymbol{x}}$: $1 + \frac{1}{3} - 1 = \frac{1}{3}$.

- **Iteration 3**

  Adding the cover inequality $x_1 + x_3 \leq 1$, we obtain the following LP, denoted LP_2:

$$\begin{array}{rrcrcrcrcl}
\max & 31\,x_1 & + & 13\,x_2 & + & 5\,x_3 & + & 9\,x_4 & & \\
\text{subject to} & 8\,x_1 & + & 6\,x_2 & + & 3\,x_3 & + & 8\,x_4 & \leq & 9 \\
& x_1 & + & x_2 & & & & & \leq & 1 \\
& x_1 & & & + & x_3 & & & \leq & 1 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}$$

An optimal solution of LP_2 is $\tilde{\boldsymbol{x}} = \left(1, 0, 0, \frac{1}{8}\right)$, with value $z(\mathrm{LP}_2) = \frac{257}{8}$.

The costs are $q = \left(0, 1, 1, \frac{7}{8}\right)$, the constant term is $n - \sum_{j \in \mathscr{I}} \tilde{x}_j = \frac{23}{8}$, and the complementary problem is:

$$\begin{array}{rrcrcrcrcl} \max & & & y_2 & + & y_3 & + & \frac{7}{8}\, y_4 & & \\ \text{subject to} & 8\, y_1 & + & 6\, y_2 & + & 3\, y_3 & + & 8\, y_4 & \leq & 15 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

The maximal subsets of indices that satisfy the constraint with right-hand side 15 are: $\{1,2\}$ (value 1), $\{1,3\}$ (value 1), $\{2,3\}$ (value 2), $\{2,4\}$ (value $\frac{15}{8}$), $\{3,4\}$ (value $\frac{15}{8}$). An optimal solution of $\mathrm{ILP}^2_{\mathrm{CSP}}$ is $\tilde{y}_1 = 0$, $\tilde{y}_2 = 1$, $\tilde{y}_3 = 1$, $\tilde{y}_4 = 0$, with value $z(\mathrm{ILP}^2_{\mathrm{CSP}}) = 2$:

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}}) = \frac{23}{8} - 2 = \frac{7}{8} < 1.$$

Accordingly, $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\} = \{1, 4\}$ and the cover inequality

$$x_1 + x_4 \leq 1$$

is maximally violated by $\tilde{\boldsymbol{x}}$: $1 + \frac{1}{8} - 1 = \frac{1}{8}$.

- **Iteration 4**

  Adding the cover inequality $x_1 + x_4 \leq 1$, we obtain the following LP, denoted LP_3:

$$\begin{array}{rrcrcrcrcl} \max & 31\, x_1 & + & 13\, x_2 & + & 5\, x_3 & + & 9\, x_4 & & \\ \text{subject to} & 8\, x_1 & + & 6\, x_2 & + & 3\, x_3 & + & 8\, x_4 & \leq & 9 \\ & x_1 & + & x_2 & & & & & \leq & 1 \\ & x_1 & & & + & x_3 & & & \leq & 1 \\ & x_1 & & & & & + & x_4 & \leq & 1 \\ & x_1 & & & & & & & \leq & 1 \\ & & & x_2 & & & & & \leq & 1 \\ & & & & & x_3 & & & \leq & 1 \\ & & & & & & & x_4 & \leq & 1 \\ & x_1, & & x_2, & & x_3, & & x_4 & \geq & 0 \end{array}$$

  An optimal solution of LP_3 is $\tilde{\boldsymbol{x}} = (1, 0, 0, 0)$, with value $z(\mathrm{LP}_3) = 31$.

  The costs are $q = (0, 1, 1, 1)$, the constant term is $n - \sum_{j \in \mathscr{I}} \tilde{x}_j = 3$, and the complementary problem is:

$$\begin{array}{rrcrcrcrcl} \max & & & y_2 & + & y_3 & + & y_4 & & \\ \text{subject to} & 8\, y_1 & + & 6\, y_2 & + & 3\, y_3 & + & 8\, y_4 & \leq & 15 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

The maximal subsets of indices that satisfy the constraint with right-hand side 15 are: $\{1,2\}$ (value 1), $\{1,3\}$ (value 1), $\{2,3\}$ (value 2), $\{2,4\}$ (value 2), $\{3,4\}$ (value 2). The optimal value of $\text{ILP}^2_{\text{CSP}}$ is $z(\text{ILP}^2_{\text{CSP}}) = 2$:

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\text{ILP}^2_{\text{CSP}}) = 3 - 2 = 1 \geq 1.$$

No cover inequality is violated by $\tilde{\boldsymbol{x}} = (1,0,0,0)$: the separation stops after 3 cover inequalities. The optimal value over the cover closure is

$$z(\text{LP}^{\text{cov}}) = z(\text{LP}_3) = 31 < z(\text{LP}) = \frac{199}{6},$$

and $\tilde{\boldsymbol{x}}$ is **integer**, hence it is also optimal for the binary problem.

Exercise 5.15 — Cover closure by separation ($P5$)

Consider a binary problem with one constraint $\sum_j w_j x_j \leq c$, with positive integer coefficients w_j and right-hand side $c = 12$, on the binary variables indexed by the set $\mathscr{I} = \{1,2,3,4\}$ of $n = 4$ variables with the following objective coefficients and constraint coefficients:

j	1	2	3	4
p_j	38	16	14	4
w_j	7	4	7	6

The associated LP is:

$$\begin{array}{rrcrcrcrcl}
\max & 38\,x_1 & + & 16\,x_2 & + & 14\,x_3 & + & 4\,x_4 & & \\
\text{subject to} & 7\,x_1 & + & 4\,x_2 & + & 7\,x_3 & + & 6\,x_4 & \leq & 12 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}$$

An optimal solution $\tilde{\boldsymbol{x}}$ of LP and its value are:

$$\tilde{x}_1 = 1, \quad \tilde{x}_2 = 1, \quad \tilde{x}_3 = \frac{1}{7}, \quad \tilde{x}_4 = 0 \qquad \text{and} \qquad z(\text{LP}) = 56.$$

Questions

1. Starting from $\tilde{\boldsymbol{x}}$, separate the cover inequalities by solving the complementary problem $\text{ILP}^2_{\text{CSP}}$ to optimality. Add the maximally violated cover inequality to the LP, solve it again, and repeat until no cover inequality is violated. If $\text{ILP}^2_{\text{CSP}}$ has several optimal solutions, choose the one that selects the largest number of variables.

2. What is the optimal value over the cover closure? Is the final solution integer?

The **cover closure** is the polyhedron obtained by adding to LP all the cover inequalities $\sum_{j\in C} x_j \le |C|-1$, $C \in \mathcal{C}$. Given a point $\tilde{\boldsymbol{x}}$, let $\tilde{y}_1, \dots, \tilde{y}_n$ be an optimal solution of the complementary problem

$$(\mathrm{ILP}^2_{\mathrm{CSP}}) \qquad \max \Big\{ \sum_{j\in\mathscr{I}} (1-\tilde{x}_j)\, y_j : \sum_{j\in\mathscr{I}} w_j\, y_j \le \sum_{j\in\mathscr{I}} w_j - (c+1),\ y_j \in \{0,1\}\ \forall j \in \mathscr{I} \Big\}.$$

If $n - \sum_{j\in\mathscr{I}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}}) < 1$, the cover $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\}$ is associated with the cover inequality maximally violated by $\tilde{\boldsymbol{x}}$; otherwise, no cover inequality is violated by $\tilde{\boldsymbol{x}}$.

Solution

Separation of the cover inequalities with the complementary problem; in all the iterations, the right-hand side of $\mathrm{ILP}^2_{\mathrm{CSP}}$ is $\sum_{j\in\mathscr{I}} w_j - (c+1) = 24 - 13 = 11$.

- **Iteration 1**
  The optimal solution of LP is $\tilde{\boldsymbol{x}} = \left(1, 1, \frac{1}{7}, 0\right)$, with value $z(\mathrm{LP}) = 56$.
  The costs are $q = \left(0, 0, \frac{6}{7}, 1\right)$, the constant term is $n - \sum_{j\in\mathscr{I}} \tilde{x}_j = \frac{13}{7}$, and the complementary problem is:

$$\begin{array}{lccccccccc} \max & & & & & \frac{6}{7}\, y_3 & + & y_4 & & \\ \text{subject to} & 7\, y_1 & + & 4\, y_2 & + & 7\, y_3 & + & 6\, y_4 & \le & 11 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0,1\} \end{array}$$

  The maximal subsets of indices that satisfy the constraint with right-hand side 11 are: $\{1,2\}$ (value 0), $\{2,3\}$ (value $\frac{6}{7}$), $\{2,4\}$ (value 1). An optimal solution of $\mathrm{ILP}^2_{\mathrm{CSP}}$ is $\tilde{y}_1 = 0,\ \tilde{y}_2 = 1,\ \tilde{y}_3 = 0,\ \tilde{y}_4 = 1$, with value $z(\mathrm{ILP}^2_{\mathrm{CSP}}) = 1$:

$$n - \sum_{j\in\mathscr{I}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}}) = \frac{13}{7} - 1 = \frac{6}{7} < 1.$$

  Accordingly, $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\} = \{1,3\}$ and the cover inequality

$$x_1 + x_3 \le 1$$

  is maximally violated by $\tilde{\boldsymbol{x}}$: $1 + \frac{1}{7} - 1 = \frac{1}{7}$.

- **Iteration 2**
  Adding the cover inequality $x_1 + x_3 \le 1$, we obtain the following LP, denoted LP_1:

$$
\begin{array}{rrcrcrcrcl}
\max & 38\,x_1 & + & 16\,x_2 & + & 14\,x_3 & + & 4\,x_4 & & \\
\text{subject to} & 7\,x_1 & + & 4\,x_2 & + & 7\,x_3 & + & 6\,x_4 & \leq & 12 \\
& x_1 & & & + & x_3 & & & \leq & 1 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}
$$

An optimal solution of LP_1 is $\tilde{\boldsymbol{x}} = \left(1, 1, 0, \frac{1}{6}\right)$, with value $z(\mathrm{LP}_1) = \frac{164}{3}$.

The costs are $q = \left(0, 0, 1, \frac{5}{6}\right)$, the constant term is $n - \sum_{j \in \mathscr{I}} \tilde{x}_j = \frac{11}{6}$, and the complementary problem is:

$$
\begin{array}{rrcrcrcrcl}
\max & & & & & y_3 & + & \frac{5}{6}\,y_4 & & \\
\text{subject to} & 7\,y_1 & + & 4\,y_2 & + & 7\,y_3 & + & 6\,y_4 & \leq & 11 \\
& y_1, & & y_2, & & y_3, & & y_4 & \in & \{0, 1\}
\end{array}
$$

The maximal subsets of indices that satisfy the constraint with right-hand side 11 are: $\{1, 2\}$ (value 0), $\{2, 3\}$ (value 1), $\{2, 4\}$ (value $\frac{5}{6}$). An optimal solution of $\mathrm{ILP}^2_{\mathrm{CSP}}$ is $\tilde{y}_1 = 0,\ \tilde{y}_2 = 1,\ \tilde{y}_3 = 1,\ \tilde{y}_4 = 0$, with value $z(\mathrm{ILP}^2_{\mathrm{CSP}}) = 1$:

$$
n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}}) = \frac{11}{6} - 1 = \frac{5}{6} < 1.
$$

Accordingly, $\tilde{C} = \{j \in \mathscr{I} : \tilde{y}_j = 0\} = \{1, 4\}$ and the cover inequality

$$
x_1 + x_4 \leq 1
$$

is maximally violated by $\tilde{\boldsymbol{x}}$: $1 + \frac{1}{6} - 1 = \frac{1}{6}$.

- **Iteration 3**
  Adding the cover inequality $x_1 + x_4 \leq 1$, we obtain the following LP, denoted LP_2:

$$
\begin{array}{rrcrcrcrcl}
\max & 38\,x_1 & + & 16\,x_2 & + & 14\,x_3 & + & 4\,x_4 & & \\
\text{subject to} & 7\,x_1 & + & 4\,x_2 & + & 7\,x_3 & + & 6\,x_4 & \leq & 12 \\
& x_1 & & & + & x_3 & & & \leq & 1 \\
& x_1 & & & & & + & x_4 & \leq & 1 \\
& x_1 & & & & & & & \leq & 1 \\
& & & x_2 & & & & & \leq & 1 \\
& & & & & x_3 & & & \leq & 1 \\
& & & & & & & x_4 & \leq & 1 \\
& x_1, & & x_2, & & x_3, & & x_4 & \geq & 0
\end{array}
$$

An optimal solution of LP_2 is $\tilde{\boldsymbol{x}} = (1, 1, 0, 0)$, with value $z(\mathrm{LP}_2) = 54$.

The costs are $q = (0, 0, 1, 1)$, the constant term is $n - \sum_{j \in \mathscr{I}} \tilde{x}_j = 2$, and the complementary problem is:

$$\begin{array}{rccccccccc} \max & & & & & & y_3 & + & y_4 & \\ \text{subject to} & 7\,y_1 & + & 4\,y_2 & + & 7\,y_3 & + & 6\,y_4 & \leq & 11 \\ & y_1, & & y_2, & & y_3, & & y_4 & \in & \{0, 1\} \end{array}$$

The maximal subsets of indices that satisfy the constraint with right-hand side 11 are: $\{1, 2\}$ (value 0), $\{2, 3\}$ (value 1), $\{2, 4\}$ (value 1). The optimal value of $\mathrm{ILP}^2_{\mathrm{CSP}}$ is $z(\mathrm{ILP}^2_{\mathrm{CSP}}) = 1$:

$$n - \sum_{j \in \mathscr{I}} \tilde{x}_j - z(\mathrm{ILP}^2_{\mathrm{CSP}}) = 2 - 1 = 1 \geq 1.$$

No cover inequality is violated by $\tilde{\boldsymbol{x}} = (1, 1, 0, 0)$: the separation stops after 2 cover inequalities. The optimal value over the cover closure is

$$z(\mathrm{LP}^{\mathrm{cov}}) = z(\mathrm{LP}_2) = 54 < z(\mathrm{LP}) = 56,$$

and $\tilde{\boldsymbol{x}}$ is **integer**, hence it is also optimal for the binary problem.